ธนาคารแห่งประเทศไทย
BANK OF THAILAND

MONETARY POLICY REPORT

March

2019



Monetary Policy Report

The *Monetary Policy Report* is prepared quarterly by staff of the Bank of Thailand with the approval of the Monetary Policy Committee (MPC). It serves two purposes: (1) to communicate to the public the MPC's consideration and rationales for the conduct of monetary policy, and (2) to present the latest set of economic and inflation forecasts, based on which the monetary policy decisions were made.

The Monetary Policy Committee
March 2019

Mr. Veerathai Santiprabhob	Chairman
Mr. Mathee Supapongse	Vice Chairman
Mr. Paiboon Kittisrikangwan	Member
Mr. Sethaput Suthiwart-Narueput	Member
Mr. Kanit Sangsubhan	Member
Mr. Subhak Siwaraksa	Member
Mr. Somchai Jitsuchon	Member

Monetary Policy in Thailand

Monetary Policy Committee

Under the Bank of Thailand Act, the Monetary Policy Committee (MPC) comprises the governor and two deputy governors, as well as four distinguished external members representing various sectors of the economy, with the aim of ensuring that monetary policy decisions are effective and transparent.

Monetary Policy Objective

The MPC sets monetary policy to promote the objective of supporting sustainable and full potential economic growth, without causing inflationary problems or economic and financial imbalances or bubbles.

Monetary Policy Target

The Cabinet approved the annual average headline inflation target of 2.5 $\pm$ 1.5 percent as the target for the medium term and for 2019. The inflation target is to assure the general public that the MPC will take necessary policy actions to return headline inflation to the target within an appropriate time horizon without jeopardizing growth and macro-financial stability. In the event that headline inflation deviates from the target, the MPC shall explain the reasons behind the target breach to the Minister of Finance and the public, together with measures taken and estimated time to bring inflation back to the target.

Monetary Policy Instrument

The MPC utilizes the 1-day bilateral repurchase transaction rate as the policy interest rate to signal the monetary policy stance.

Evaluation of Economic Conditions and Forecasts

The Bank of Thailand takes into account information from all sources, the macroeconomic model, data from each economic sector, as well as surveys of large enterprises, together with small and medium-sized enterprises from all over the country, and various financial institutions to ensure that economic evaluations and forecasts are accurate and cover all aspects, both at the macro and micro levels.

Monetary Policy Communication

Recognizing the importance of monetary policy communication to the public, the MPC employs various channels of communication, both in Thai and English, such as (1) organizing a press statement at 14:00 on the day of the Committee meeting, (2) publishing edited minutes of the MPC meeting two weeks after the meeting, and (3) publishing the *Monetary Policy Report* every quarter.

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Executive Summary

Monetary Policy Conduct in the First Quarter of 2019

In the Committee's view, the Thai economy was expected to expand around its potential, on the back of domestic demand, despite at a slower pace than previously assessed due to a slowdown in external demand. Headline inflation was projected to be in line with the previous forecast. Core inflation would be slightly lower than previously assessed while still gradually trending up. However, there were pockets of risks in the financial system that might pose vulnerabilities to financial stability in the future. **The Committee weighed various factors in determining the most appropriate course of monetary policy and voted 4 to 2 at the meeting on February 6, 2019 to maintain the policy rate at 1.75 percent with one committee member absent. At the meeting on March 20, 2019, the Committee unanimously voted to maintain the policy rate at 1.75 percent.** The Committee viewed that the current accommodative monetary policy stance contributed to the continuation of economic growth and was appropriate given the inflation target. In addition, given heightened global and domestic uncertainties in the current period, the Committee thus voted to keep the policy rate unchanged to assess the clarity of impacts from such uncertainties. Furthermore, there remained a need to address risks to financial stability through a combination of tools, including an appropriate policy rate as well as microprudential and macroprudential measures.

Looking ahead, the policy rate increase would be gradual and follow a data-dependent approach. The Committee would closely monitor developments of economic growth, inflation, and financial stability, together with associated risks, in deliberating appropriate monetary policy in the period ahead.

Assessment of the Economic and Financial Outlook as the Basis for Policy Formulation

1. Global Economy

The global economy was expected to expand at a slower rate, but still close to its potential. Private consumption of most trading partner economies continued to expand, consistent with strong labor market conditions and government support in some countries. Nonetheless, economic outturns in the fourth quarter of 2018 and developments in the first quarter of 2019 reflected that economic growth was below expectations in many economies. This was due to the impact from global trade slowdown as well as country-specific factors. Such factors included a temporary impact from the U.S. government shutdown, hampered confidence in Europe following prolonged political issues, and uncertainties surrounding trade protectionist measures between the U.S. and China. **The Committee thus revised down the growth projection of Thailand's trading partner economies to be 3.2 percent for both 2019 and 2020. However, there remained possibilities that growth in the trading partner economies would underperform the baseline projection** due to uncertainties pertaining to trade negotiations between the U.S. and China as well as Brexit negotiations. Other risks that still warranted monitoring included uncertainties regarding the impact of trade protectionist measures on the global value chain, geopolitical risks, and China's financial stability concerns.

Most central banks maintained their monetary policy stance and followed a data-dependent approach, while some central banks started to signal a dovish stance toward policy normalization. The U.S. Federal Reserve (Fed) was expected to raise the policy rate once in 2019[1/] from the previous anticipation of two rate hikes. The European Central Bank (ECB) ended new bond purchases at the end of 2018 according to its announced plan but continued rolling over matured bonds. The ECB was expected to keep the policy rate on hold throughout 2019 and implement a plan to increase liquidity in the financial system through targeted longer-term refinancing operations (TLTROs). Meanwhile, the Bank of Japan (BOJ)

[1/] This was an assessment as of 19 March 2019 before the Federal Open Market Committee (FOMC) meeting at 20 March 2019, with the tone of the statement dovish than the market expected.

would maintain both short- and long-term interest rate targets for some period. Many central banks in the region, including the Bangko Sentral ng Pilipinas and Bank Indonesia, maintained their policy rates after having continuously raised the rate in 2018. Other central banks such as the Bank of Korea, Bank Negara Malaysia, and the Reserve Bank of Australia kept their policy rates on hold due to the outlook of economic slowdown, declining inflation, and uncertainties on trade protectionist measures.

Emerging markets (EMs) experienced capital inflows after risks in the global financial market declined. This was due to the Fed's dovish stance regarding the monetary policy communications, improvements in trade negotiations between the U.S. and China, as well as lower risks of the no-deal Brexit during the beginning of the year. However, geopolitical risks affected investor confidence in certain periods. Looking ahead, the global financial market would remain highly volatile. International capital flows both into and out of EMs could fluctuate depending on monetary policy directions of advanced economies, prolonged uncertainties pertaining to U.S.-China trade protectionist measures, uncertainties regarding Brexit negotiations, political developments in the euro area as well as geopolitical risks.

2. Financial Conditions and Financial Stability

Thailand's financial conditions remained accommodative. The new loan rate remained stable at a low level despite the gradual increase of short-term government bond yields in line with the policy rate. Meanwhile, medium- and long-term government bond yields fluctuated due to both domestic and external factors. Private credit expansion continued in both business and household sectors. Business sectors continued to raise fund through both bond and equity markets. The Thai baht appreciated against the U.S. dollar at the beginning of the quarter, consistent with other regional currencies, due to (1) a weaker U.S. dollar after the Fed's dovish stance regarding the monetary policy communications, (2) a lower possibility of the no-deal Brexit during the beginning of the year, and (3) improvements in U.S.-China trade negotiations. The baht appreciated to a larger extent as compared with other regional currencies given Thailand's large current account surplus and investors associating the baht with low risks. In late February 2019, the baht depreciated against the U.S. dollar due to better-than-expected outturns of some U.S. economic figures. Moreover, investors were concerned about geopolitical risks between the U.S. and North Korea as well as the general election in Thailand. Meanwhile, the real effective exchange rate (REER) appreciated.

There were some risks in the financial system that could pose vulnerabilities to financial stability in the future. These risks included (1) an accumulation of household debt especially for auto and mortgage loans, (2) the search-for-yield behavior which persisted in the low interest rate environment and could lead to underpricing of risks, particularly among saving cooperatives and large corporates, and (3) risks in the property sector including uncertainties regarding Chinese demand for Thai condominiums. There remained a need to monitor adjustments in the property market and mortgage loans after the revised macroprudential measure on mortgage loans were to be effective in April 2019.

3. Economic and Inflation Outlook

The Thai economy was projected to register slower growth at 3.8 and 3.9 percent in 2019 and 2020, respectively, down from the projection in the previous *Monetary Policy Report*. Growth drivers stemming from external demand were expected to moderate following the slowdown in trading partner economies and global trade volume. Merchandise and services export growth was thus expected to slow down. Meanwhile, domestic demand would remain a key growth driver.

Merchandise exports were expected to slow down in line with global demand. Export volume was projected to moderate in tandem with the outlook of trading partner economies and global trade volume. Looking ahead, Thailand's merchandise exports would be supported by the relocation of production base in some industries to Thailand, redirected orders from China to Thailand, and 5G technology-related infrastructure investment plans in many countries, which would benefit Thailand's exports of electronics parts in the period ahead.

Exports of services were projected to exhibit slower growth mainly due to lower spending per head, despite a higher-than-expected number of foreign tourists. The global economic slowdown could lead to lower spending per head, especially for Chinese tourists, and a shorter trip duration of European tourists. The projected number of foreign tourist arrivals was revised up to 40.4 and 42.0 million for 2019 and 2020, respectively. The upward revision was on the back of tourist confidence on improvements in Thailand's safety standards and the extended exemption of visa-on-arrival (VOA) fees until the end of April 2019.

Private consumption was expected to be well maintained despite some moderation in durable goods which had accelerated significantly in 2018. Supporting factors included a continued improvement in household income for both farm and non-farm households, improved consumer confidence, and support from government policies. However, elevated household debt would weigh on consumption. In addition, the adoption of automation in place of labor in the production process and inflation remaining at a low level would limit pressures on employers to increase wages, thus resulting in a gradual improvement in household purchasing power.

Private investment, particularly investment in machinery and equipment, would continue to expand. Supporting factors included expansion in production capacity and the relocation of production base to Thailand for hard-disk drives and other industries affected by the trade protectionist measures between the U.S. and China, which would lend support to Thai exports going forward. In addition, the private sector was expected to invest further in public-private partnership (PPP) investment projects.

Public spending would continue to drive the economy, especially investment on transportation infrastructure including the dual-track railway and mass rapid transit projects. Nevertheless, public investment was projected to grow at a slower pace than previously assessed due to delays in some state-owned enterprises' infrastructure investment projects and changes in the investment structure of some PPP projects to allow for greater investment by the private sector. Moreover, the budget allocation for replenishment of the treasury account balance for fiscal year 2020 was higher than expected, resulting in lower composition of capital and current expenditures.

Inflation was expected to trend up consistent with the previous assessment. Fresh food prices were expected to rise on account of an intensified drought and would offset some decline in core inflation following the softening in economic activity. However, demand-pull inflationary pressures would subsequently slowly rise reflecting a gradual closing of the output gap in the period ahead. **Thus, the Committee projected headline inflation to average 1.0 and 1.1 percent for 2019 and 2020, respectively, and core inflation to average 0.8 and 0.9 percent for 2019 and 2020, respectively.**

The growth projection was subject to downside risks due to (1) lower-than-expected trading partner economic growth following such as more intensified trade protectionist measures between the U.S. and China, uncertainties regarding Brexit negotiation, the U.S. import tariffs on automobiles and parts, and geopolitical risks; and (2) lower-than-expected private investment due to political uncertainties. However, **there was a possibility that the Thai economy would outperform the baseline projection** due to (1) less-than-expected slowdown of the Chinese economy due to government economic stimulus measures; (2) higher-than-expected domestic demand as a result of (2.1) sooner-than-expected implementation of government infrastructure investment projects, PPP projects, and private investment, (2.2) a quick dissolving of political uncertainty, and (3) additional government measures to support private spending. Meanwhile, **risks to the forecasts of headline and core inflation were expected to tilt downward in line with increased downside risks** to growth projections.

1. Global Economy

Major advanced economies were expected to grow at a slower rate. The U.S. economy mainly experienced a temporary impact from the government shutdown, while the euro area and Japanese economies moderated due to country-specific factors and global trade slowdown.

The U.S. economy was projected to slow down in the fourth quarter of 2018. Private consumption moderated in line with consumer confidence, which was partly a result of a sharp fall of stock indices. Meanwhile, some economic activities faced a temporary impact from the government shutdown in the first quarter of 2019. The manufacturing sector would likely slow down, after having accelerated in building up inventories prior to the increase in import tariffs between the U.S. and China. However, strong labor market (Chart 1.1), high consumer confidence, as well as personal and corporate income tax cuts would be supporting factors for domestic demand to drive the U.S. economy in the period ahead. **The euro area was projected to grow at a slower rate** because the new emission standard had a longer-than-expected impact on automobile production, low water levels of the Rhine River hindered merchandise exports from Germany, and political uncertainty in some countries, such as France and Italy, hampered confidence and economic activities. In addition, merchandise exports from the euro area decreased as global trade volume was affected from the trade protectionist measures between the U.S. and China, as well as the global economic slowdown. Nonetheless, the euro area were projected to expand on the back of private consumption, as supported by labor markets that remained robust. **The Japanese economy was expected to moderate** due to lower merchandise exports in response to global trade outlook. Going forward, the Japanese economy was expected to expand primarily on the back of domestic demand with support from the labor market, government economic stimulus measures, and continuously accommodative monetary policy. Furthermore, private consumption was expected to accelerate prior to an increase in consumption tax in October 2019.

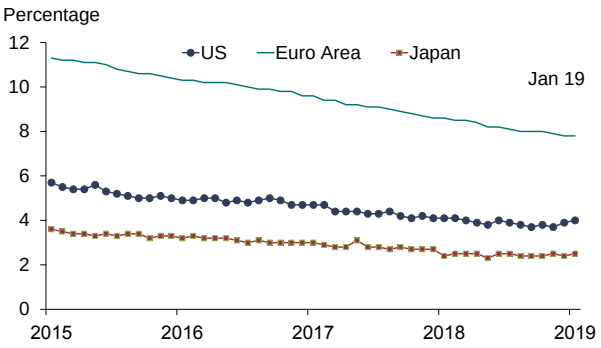
The Chinese economy slowed down due to a series of financial stability measures implemented in recent periods. Meanwhile, moderating global trade had a larger-than-expected impact on other Asian economies.

The Chinese economy was expected to slow down slightly due to a series of financial stability measures to curb debt in the economy implemented in recent periods. In addition, impacts from the U.S. trade protectionist measures on China's merchandise exports were expected to be increasingly prominent in the period ahead, particularly from measures that were already imposed. However, the Chinese government's growth-supportive measures, including accelerated infrastructure investment, tax deductions for businesses and households, and liquidity injection into the financial system by reducing the banks' reserve requirement ratio (RRR), would help alleviate some adverse impacts and support the economy to expand around 6.0-6.5 percent in 2019, consistent with the government's target.

Asian economies, excluding Japan and China, were expected to exhibit slower growth. Merchandise exports of several countries began to slow down in line with global trade volume, which was affected by the trade protectionist measures between the U.S. and China and the waning effect of front-loading exports ahead of the full implementation of tariffs

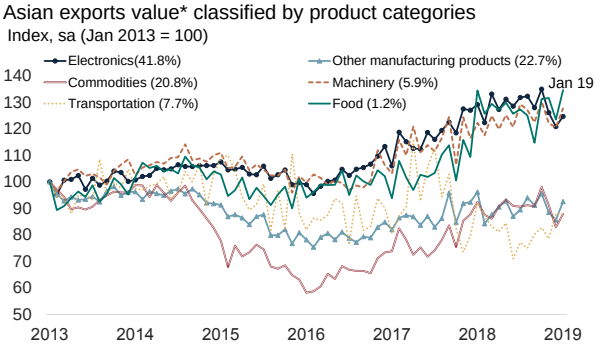
imposed by the U.S. and China (Chart 1.2). Looking ahead, merchandise exports would likely slow down in line with global trade volume. Nonetheless, Asian economies were expected to expand on the back of domestic demand, as supported by strong labor markets, an increase in minimum wages in several countries, as well as fiscal stimulus through infrastructure investment projects along with measures to support low-income earners in some countries.

Chart 1.1 Unemployment rate continued to fall in the US, the euro area, and Japan



Source: Bloomberg

Chart 1.2 Asian exports started slowing down after the waning effect of front-loading exports ahead of the full implementation of the U.S.-China tariffs



Note: *Asian exports include Hong Kong, Taiwan, South Korea, Malaysia and Singapore.
( ) denotes share of total exports in 2017
Commodity-related products include crude oil, metals, chemicals, rubber, and vegetable oil.
Other manufacturing products include textile, papers, furniture, footwear and miscellaneous

Source: CEIC

The growth projection for Thailand's trading partners was revised downward, but remained close to its potential.

Going forward, Thailand's trading partner economies were expected to register slower growth than previously assessed in the previous *Monetary Policy Report*. The downward revision was due to lower-than-expected growth outturns during the fourth quarter of 2018 and the first quarter of 2019 in several countries, which were affected by moderating global trade and aforementioned country-specific factors. However, most trading partner economies would continue expanding around their potential levels, with private consumption and government spending as key growth drivers. **The committee, thus, revised down the growth projection for Thailand's trading partners from 3.4 percent to 3.2 percent in 2019 and expected growth in 2020 to be around this year's growth of 3.2 percent** (Table 1.1).

There remained possibilities that growth of trading partner economies would underperform the baseline projection due to prolonged uncertainties pertaining to trade negotiations between the U.S. and China, despite some improvement after the U.S. announced to delay the increase in tariffs on imported goods from China indefinitely, as well as uncertainties surrounding Brexit negotiations. Other risks that still warranted monitoring included, first, uncertainties regarding impacts from trade protectionist measures on the global supply chain and business sentiment. Second, geopolitical risks remained, particularly the denuclearization agreement between the U.S. and North Korea which had yet to materialize. Moreover, recurring tensions in other regions could heighten volatility in the global financial and commodity markets, especially through oil prices, and would, in turn, affect the real economy. Moreover, there remained political uncertainties in Europe, for instance, the unrest in France and the election in Spain. Third, economic and financial concerns in China still

persisted, although the Chinese government implemented a series of financial stability measures to address these issues to some extent. Nonetheless, there remained challenges in balancing between promoting economic growth and preserving financial stability.

Table 1.1 Assumption on trading partner growth

Annual change (%YoY)	Weight (%)	2018*	2019	2020
United States	14.9	2.9 (2.9)	2.5 (2.6)	1.9
Euro area	10.0	1.8 (1.9)	1.1 (1.6)	1.5
Japan	13.6	0.8 (0.8)	0.9 (1.0)	0.4
China	15.7	6.6 (6.6)	6.2 (6.2)	6.0
Asia (excluding Japan and China)**	37.4	4.1 (4.2)	3.7 (3.9)	3.7
Total***	100	3.6 (3.6)	3.2 (3.4)	3.2

Note: *Outturn
**Weighted by a share of Thailand's total exports to 7 trading partners in 2014, namely Singapore (6.5%), Hong Kong (7.9%), Malaysia (8.0%), Taiwan (2.5%), Indonesia (5.9%), South Korea (2.8%), and the Philippines (3.7%)
***Weighted by a share of exports from Thailand to 13 trading partners in 2014 (including the United Kingdom and Australia)
( ) as reported in Monetary Policy Report, Demcember 2018

Most central banks maintained their monetary policy stance and followed a data-dependent approach, while some central banks signalled a more dovish stance toward policy normalization.

The U.S. Federal Reserve (Fed) raised the policy rate 4 times in 2018 on account of a robust economic growth and continuously tightening labor market. In 2019, the Fed was expected to be more dovish toward monetary policy normalization, raising the policy rate once instead of two times as previously assessed[2/]. The dovish stance was on account of softening inflation prospects, moderating global economic outlook, and heightened volatility in the financial market. Going forward, the Fed's monetary policy would mainly follow a data-dependent approach. Meanwhile, the European Central Bank (ECB) communicated its monetary policy stance in a more dovish manner. The ECB ended new bond purchases at the end of 2018 as previously announced, but continued rolling over matured bonds. The ECB was expected to keep the policy rate on hold throughout 2019, instead of until the second half of 2019 as previously expected. Moreover, the ECB would also implement a plan to increase liquidity in the financial system through targeted longer-term refinancing operations (TLTROs), starting from September 2019 until March 2021. Meanwhile, the Bank of Japan (BOJ) would maintain both short- and long-term interst rate targets for some period. Many central banks in the region, including the Bangko Sentral ng Pilipinas (BSP) and Bank Indonesia (BI), began to maintain their policy rates in the first quarter of 2019 after having continuously raised the policy rate in 2018. Other central banks, such as the Bank of Korea (BOK), Bank Negara Malaysia (BNM) and the Reserve Bank of Australia (RBA), kept their policy rate on hold due to the slowdown of economic growth outlook since 2018, declining inflation and uncertainties on trade protectionist measures.

[2/] This assessment was made on 19 March 2019 before the Federal Open Market Committee (FOMC) meeting at 20 March 2019, with the tone of the statement being more dovish than the market expected.

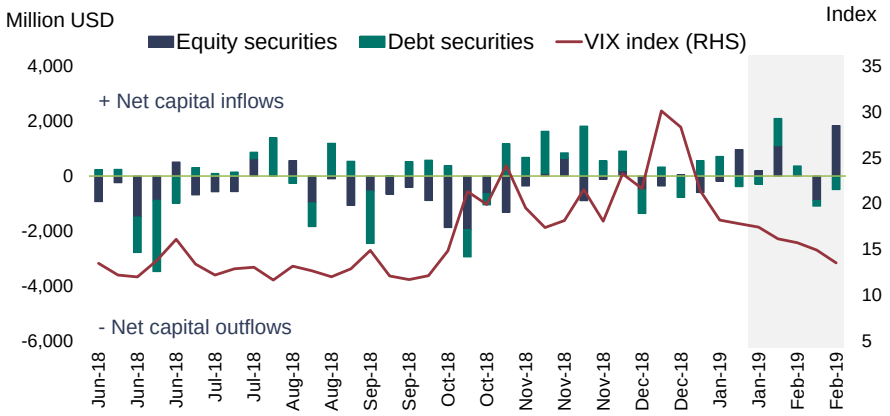
Emerging markets (EMs) experienced capital inflows following the Fed's dovish stance in the monetary policy communications, easing concerns over the U.S.-China trade protectionist measures, as well as lower risks of the no-deal Brexit at the beginning of 2019.

Overall risks in the global financial market declined at the beginning of 2019 due to the Fed's dovish stance regarding the monetary policy communications, improvements in trade negotiations between the U.S. and China, as well as lower risks of the no-deal Brexit, as reflected by a substantial decline in VIX Index[3/] (Chart 1.3). Investors, thus, increased their risk appetite and returned to invest continually in EM assets, particularly in equity markets, consistent with rising stock indices. However, geopolitical risks had an impact on investor confidence in some periods, such as tension between India and Pakistan and the abrupt end of the U.S.-North Korea summit.

Looking ahead, the global financial market would remain highly volatile. International capital flows, both into and out of EMs, could fluctuate depending on monetary policy directions of advanced economies, prolonged uncertainties pertaining to U.S.-China trade protectionist meaures, uncertainties regarding Brexit negotiations, political developments in the euro area, as well as geopoitical risks.

Chart 1.3 EMs experienced capital inflows at the beginning of the year due to Fed's dovish communications, easing concerns over the U.S.-China trade protectionist measures, as well as lower risks of the no-deal Brexit

Net capital inflows to EMs* (weekly) and VIX index



Note: *EMs include Thailand, Indonesia, India, South Africa, and Turkey
Sources: Bloomberg and Institutional Institute of Finance

Crude oil prices in the first quarter of 2019 declined from the previous quarter. Looking ahead, despite moderating demand for crude oil following the global economic slowdown, tightening crude oil supply would support oil prices.

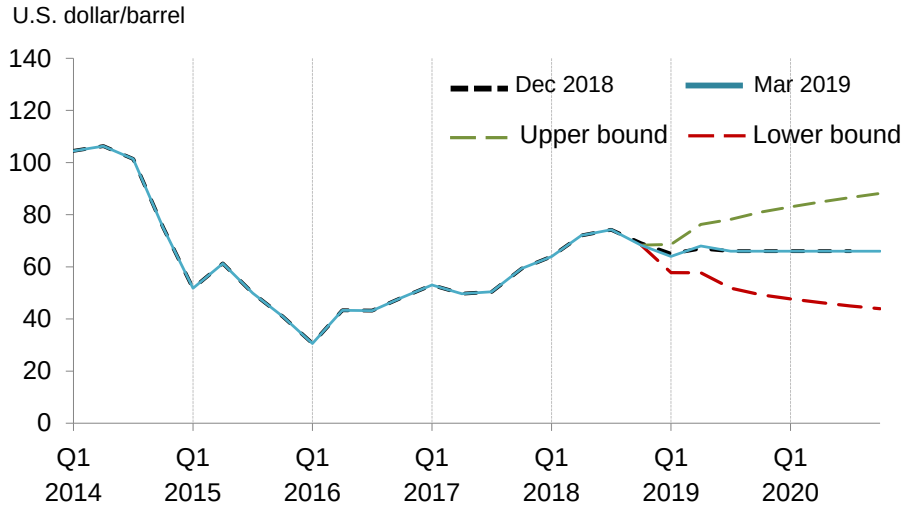
In the first quarter of 2019, the Dubai crude oil prices declined from the previous quarter. Such decline was owing to investors' concerns over a softening outlook of the global economy. Besides, the U.S. Department of Energy also expected the U.S. crude oil production to rise in 2019 and 2020, due mainly to an increase in crude oil supply by shale oil producers. However, oil prices gradually picked up since the middle of the first quarter of 2019,

[3/] VIX Index is a measure of the stock market volatility implied by S&P 500 index options

as investors projected the global oil supply to tighten in the period ahead due to several factors including (1) a lower production capacity of the OPEC members and allies according to the agreement to cut production, (2) the U.S. sanction against Venezuela, and (3) the end of sanction exemptions for some of Iran's oil exports starting in the second quarter of 2019.

The Committee thus maintained the projection for Dubai crude oil prices throughout the forecast horizon. Despite moderating demand for crude oil following the global economic slowdown, tightening oil supply would support oil prices. Therefore, the Committee maintained the projection for Dubai crude oil prices at 66.0 dollars per barrel in 2019 and 2020 (Chart 1.4). Moreover, **there were risks that oil prices would be below the baseline projection,** similar to those evaluated in the previous *Monetary Policy Report*. In the short run, crude oil prices could rise in certain periods due to uncertainties pertaining to the U.S. sanctions against Venezuela and Iran, which might have an impact on the projection of crude oil supply and investor sentiment. In the period ahead, there remained several factors which could pressure oil prices to stay below the baseline projection. These risks included a larger-than-expected global economic slowdown, a possibly faster-than-expected increase in the U.S. shale oil production, the cut in oil production by OPEC members and allies which could be lower than the agreement, as well as the enforcement of the new marine fuel regulations imposed by the International Maritime Organization (IMO) which might lower demand for high-sulpher crude oil including Dubai crude oil.

Chart 1.4 Dubai crude oil prices in 2019Q1 declined from the previous quarter. Looking ahead, despite moderating demand for crude oil following the global economic slowdown, tightening crude oil supply would support oil prices



2. The Thai Economy

2.1 Recent Developments

The Thai economy continued to gain traction in the fourth quarter of 2018, mainly driven by domestic demand while external demand gradually improved.

In the fourth quarter of 2018 the Thai economy expanded 3.7 percent from the same period last year. The key driver was private consumption which continued expanding across almost all product categories. In addition, purchasing power of farm and non-farm households improved, partly thanks to higher output of major crops such as rice, tapioca, rubber, vegetables and fruit, and government measures aiming to relieve costs of living of the lower income groups. Private investment accelerated, driven by investment in machinery and equipment, especially in vehicle and transport equipment. Meanwhile, construction investment continued to expand, partly on account of public-private partnership (PPP) for the mass rapid transit projects. Public spending growth moderated somewhat due to a slight contraction in public investment in tandem with lower disbursement in the central government's current-year and carry-over budget, partly as a result of government project review to comply with the master plans under the National Strategy. With regard to external demand, merchandise exports expanded slightly on the back of increased exports of gold and some major manufacturing products such as metal products, auto parts, and sugar. However, some other exports-products were affected by the slowdown in trading partner economies, trade protectionist measures between the U.S. and China, as well as the down cycle of electronic products. As a result, overall export growth moderated. Export of services slightly improved in line with increased tourism revenues and higher numbers of tourists figures relative to the previous quarter. Overall, the Thai economy expanded 0.8 percent in the fourth quarter of 2018 after seasonal adjustment. The annual growth for 2018 was 4.1 percent, an improvement over 4.0 percent in 2018.

In the first quarter of 2019, the Thai economy continued to gain momentum, as reflected by recent economic indicators. Key growth driver was domestic demand, as private consumption continued to expand on account of improved and more broad-based household income improvement–and government measures aiming to support low-income households. However, purchasing power of households would be constrained by elevated household debt that continued to trend up. Private investment expanded, especially investment in machinery and equipment. Public spending expanded on the back of both procurement of goods and services and compensation of civil servants, while public investment expanded in both the central government and state-owned enterprises. Merchandise exports slowed down on account of trading partners' economic slowdown and the U.S. trade protectionist measures. Export of services expanded mainly due to a higher number of Chinese, Indian, and Taiwanese tourists thanks to an extended exemption of visa-on-arrival (VOA) fees from January 13, 2019 to April 30, 2019, the opening of new flight routes from China and India, and tourist confidence on improvements in Thailand's safety standards.

Overall employment level remained high. Non-agricultural employment remained high and was more broad-based, while agricultural employment stabilized after a decline following the drought. Overall purchasing power continued to expand thanks to improvement in both farm and non-farm household income.

The economic expansion contributed to improvements in household purchasing power and high employment (Charts 2.1 and 2.2). **Employment of non-farm households** remained high and more broad-based, particularly the manufacturing sector. Employment in the service sector increased, while employment in the construction and commerce sectors remained high. The slowdown in merchandise exports since the end of 2018 had yet to significantly impact on overall employment. However, some export-related sectors such as electronics and rubber products observed overtime labor hours, partly due to industry-specific factors such as the product cycle and the adoption of automation. Overall non-farm income continued to increase across sectors and income groups, particularly the middle-income and high-income households.

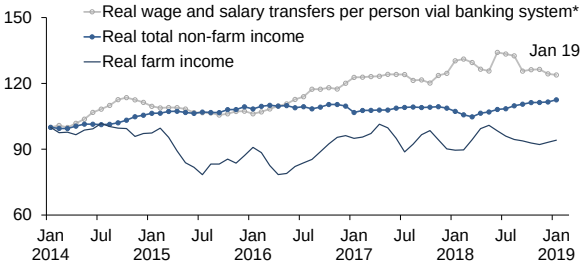
Employment of farm households remained stable after a decline owing to the drought situation mainly in the Northeast region. Despite lower output of jasmine rice following the drought, overall farm income expanded due to output increase in some crops, less contraction in crop prices, and a constant rise in rice prices. In addition, government measures aiming to assist rice, rubber, and oil palm growers helped shore up purchasing power of farm households. Looking ahead to the first half of 2019, output of some crops such as rice and fruits might reduce compared to last year. This is because there remained risk of drought due to the El Nino, which could lead to lower dam water levels than last year and low rainfalls during February-April 2019.

The favorable labor market condition in the previous periods supported purchasing power and consumption of households in various aspects, including the number of employed workers, improvements in household income, and persistently low unemployment rates. Analyzing labor market conditions could utilize a multitude of indicators to provide a comprehensive picture, for example, labor market sentiment, employer behavior, labor potential. This would help reflect structural problems in Thai labor market and lead to the appropriate formulation of labor and economic policies going forward. (Box: *Implications of low unemployment rate in Thailand*)

Chart 2.1 Overall purchasing power continued to improve

Household income indicators

Index, seasonally adjusted (3-month moving average)
(Jan 2014 = 100)

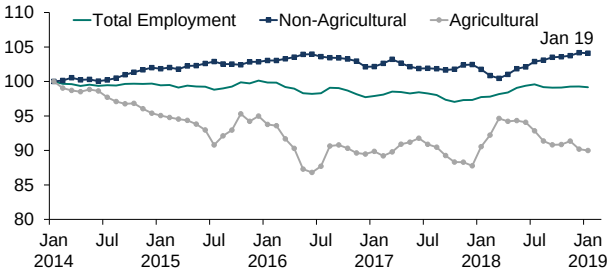


Note: *wage and salary transfer transactions are calculated from 2 databases:
(1) Commercial banks reporting transactions to Bank of Thailand database which covers 90% of all retail transfer transactions and
(2) Interbank Transaction Management and Exchange (ITMX) database which covers 10% of all retail transfer transactions.
Sources: Bank of Thailand, Office of Agricultural Economics, National Statistical Office, Ministry of Commerce, calculations by Bank of Thailand

Chart 2.2 Overall employment remained high

Employment indicators

Index, seasonally adjusted (3-month moving average)
(Jan 2014 = 100)

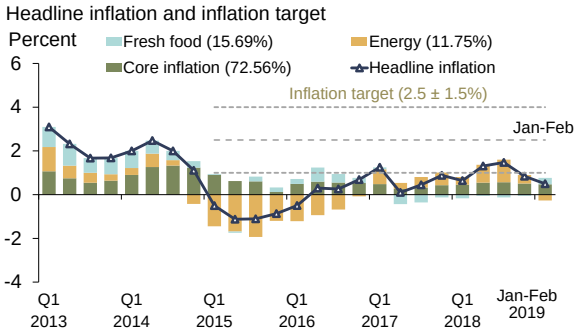


Source: National Statistical Office

Headline inflation declined mainly due to the fall in energy prices and core inflation.

Headline inflation averaged at 0.50 percent over the first two months of the first quarter of 2019, down from 0.84 percent in the previous quarter (Chart 2.3). The drop was due to declines in energy prices as domestic retail oil prices fell in line with global oil prices. On the other hand, fresh food prices increased, mainly due to increased prices of pork and eggs as excess supply abated. Vegetable and fruit prices declined to a lesser degree as there were smaller supplies in the market.

Chart 2.3 Headline inflation decreased from the previous quarter mainly due to decline in energy prices and core inflation

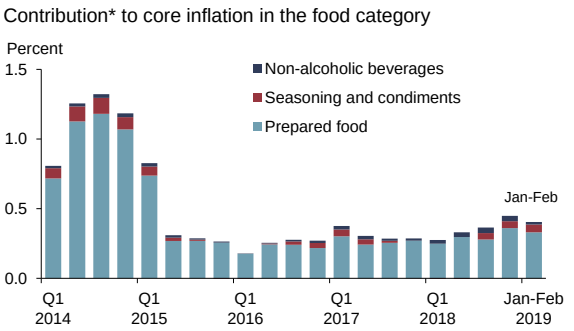


Note: ( ) denotes share in inflation baskets
Source: Ministry of Commerce, calculations by Bank of Thailand

Core inflation averaged at 0.65 percent over the first two months of the first quarter of 2019, down from 0.71 percent in the previous quarter. Core inflation in the food category declined (Chart 2.4) due to slower increases in prices of processed foods and non-alcoholic beverages (Chart 2.5). Core inflation in the non-food categories declined mainly due to a slower increase in house rents. In other categories, prices slowly increased as domestic demand gradually rose. Moreover, structural factors encompassing rising e-commerce trends, intensified price competition, and lower production costs from higher production efficiency weighed down core inflation in non-food categories.

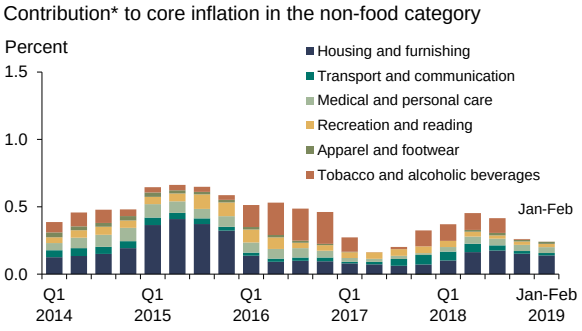
Short-term (one-year ahead) inflation expectations according to the survey of businesses in February 2019 stood at 1.7 percent, down from 1.9 percent in the previous quarter. Inflation expectations of professional forecasters in March 2019 stood at 1.5 percent, up slightly from 1.3 percent in the previous quarter. **Long-term (five-year ahead) inflation expectations** according to the survey of professional forecasters in October 2018 stood at 1.8, down from the previous survey (April 2018).

Chart 2.4 Core inflation in the food category (28% of core inflation) declined mainly on account of prepared food and non-alcoholic beverages prices



Note: *Contributions to core inflation decompose core inflation into an inflation rate of each component within the core inflation basket, weighted by its corresponding share in the basket.
Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

Chart 2.5 Core inflation in the non-food category (72% of core inflation) decreased slightly, as house rents rose at a slower rate



Note: *Contributions to core inflation decompose core inflation into an inflation rate of each component within the core inflation basket, weighted by its corresponding share in the basket.
Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

Table 2.1 Inflation

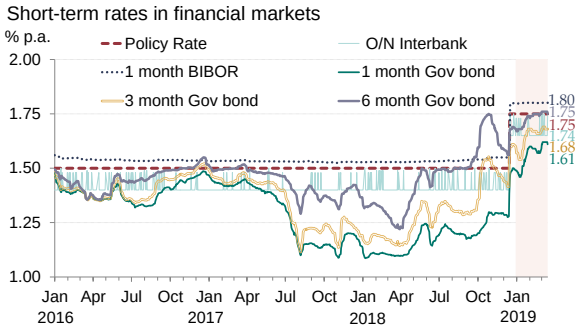
Annual percentage change	2017				2018				2019
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Jan-Feb
Headline Consumer Price Index (Headline CPI)	1.25	0.10	0.45	0.88	0.64	1.31	1.47	0.84	0.50
Core Consumer Price Index (Core CPI)	0.66	0.47	0.49	0.61	0.61	0.76	0.78	0.71	0.65
Raw food	0.61	-2.99	-2.25	-0.80	-1.04	-0.35	-0.82	-0.35	1.89
Energy	6.69	2.67	4.86	5.24	3.01	7.30	9.11	3.39	-2.21

Source: Bureau of Trade and Economic Indices, Ministry of Commerce

Short-term money market rates and short-term government bond yields gradually rose in line with the policy rate. Medium-term and long-term government bond yields fluctuated owing to external and domestic factors.

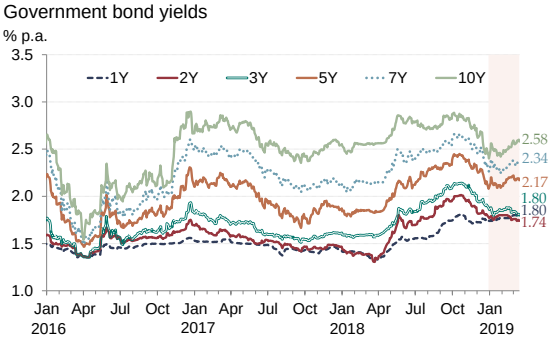
In the first quarter of 2019, **short-term money market rates** rose in line with the policy rate.**Short-term government bond yields** rose gradually (Chart 2.6) especially at the end of January, on account of higher supply of the Bank of Thailand's 14-day bonds. **Yields on two- to three-year government bonds** were stable in the beginning of the quarter before falling slightly at the end of February (Chart 2.7) partly due to higher demand by investors following the Ministry of Finance's announcement of the first bond switching in FY2019. The source bonds accepted by the Ministry of Finance were due to mature in around 3 years. **Yields on five-year and longer-term government bonds** rose gradually mainly on account of net sales by foreign investors. At times, these bond yields fluctuated in tandem with U.S. government bonds yields, influenced by positive developments in the U.S.-China trade negotiations and the outlook of global economic slowdown, and more dovish stance of monetary policy communicated by the Fed and the ECB .

Chart 2.6 Short-term money market rates and short-term government bond yields gradually rose in line with the policy rate



Sources: Bank of Thailand and Thai Bond Market Association (Thai BMA) (data as of 19 March 2019)

Chart 2.7 Yields on two-to-three-year government bonds fell slightly. Yields on over five-year government bonds increased gradually despite fluctuation in some periods



Source: Thai Bond Market Association (Thai BMA) (data as of 19 March 2019)

Corporate bond yields slightly edged up in line with government bond yields in the first quarter of 2019. Meanwhile, credit spread[4/] was stable after rising in the previous quarter. **Cost of financing through commercial banks**, as represented in the new loan rate (NLR) [5/] remained stable at about 4.0 percent (Chart 2.8). Meanwhile, reference loan rates of commercial banks

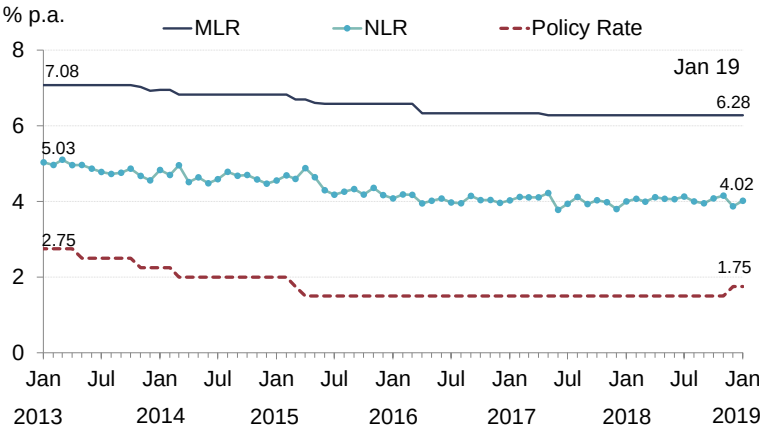
[4/] A credit spread is the difference between corporate and government bond yields with the same tenure, reflecting an assessment on corporate bond issuers' default risks.

[5/] NLR is calculated based on a weighted average of interest rates for new loan contracts extended by 14 Thai commercial banks (excluding consumer loans and loans to financial intermediaries). The data covers loans of value of 20 million baht or higher for all purposes and terms and includes both secured and non-secured loans. Moreover, interest rates used in the calculation refer to the mid-rate between the lowest and the highest rates in each loan contract.

remained mostly unchanged from the previous quarter, with the exception of a slight increase in one small bank's minimum retail rate.

Chart 2.8 New Loan Rate (NLR) stabilized at low level

New Loan Rate



Source: Bank of Thailand (data as of January 2019)

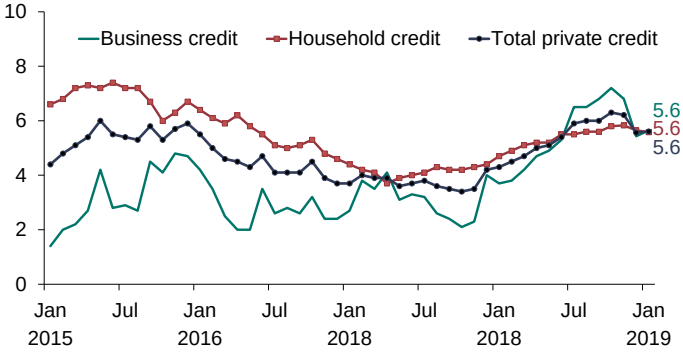
Private credit extended to both businesses and households continued to expand in line with the economic expansion.

Private credit[6/] in January 2019 expanded 5.6 percent, the same rate as in the fourth quarter of 2018. Overall business credit expanded, which was driven by loans extended to large corporates, particularly in the commerce and service sectors, and SMEs which had relatively large credit lines such as real estate businesses. Household credit continued to rise across all loan purposes in line with strong private consumption growth (Chart 2.9).

Chart 2.9 Private credit to both businesses and households continued to expand

Growth of private credit

Percentage change from the same period last year



Note: 1) Private credit includes credit to other depository corporations (ODCs) namely commercial banks, specialized financial institutions, finance companies, saving cooperatives, and money market mutual funds

2) The data of ODCs credit outstanding to business and household sectors since January 2015 are revised following the improvement of processing system for more accurate and comprehensive ODCs credits data

Source: Bank of Thailand

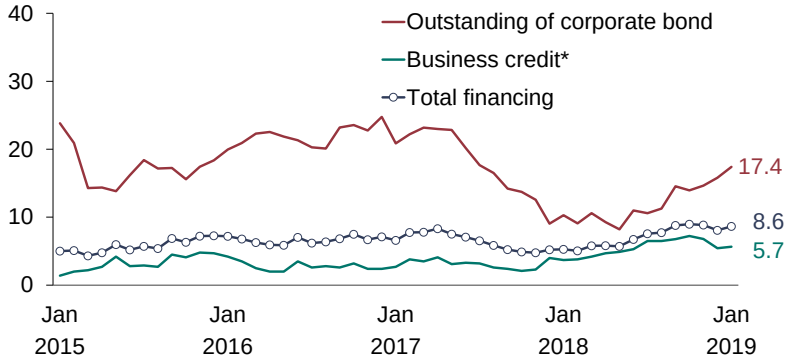
6/ Outstanding credit of other depository corporations (ODCs), namely commercial banks, specialized financial institutions, finance companies, savings cooperatives, and market mutual funds.

The net issuance of corporate bonds continued to increase in the fourth quarter of 2018 and in January 2019 relative to the previous quarter, mainly driven by corporate funding in the IT and telecommunication, retail and wholesale trade, and energy sectors. Overall, net corporate bond outstanding rose 17.4 percent from the same period of last year (Chart 2.10). **Funding through the equity market** continued to increase in the fourth quarter of 2018 and in January 2019, particularly in the IT and telecommunication, real estate, and electronic component sectors.

Chart 2.10 Overall financing continued to expand

Growth of corporate bond outstanding and business credit

Percentage change from the same period last year



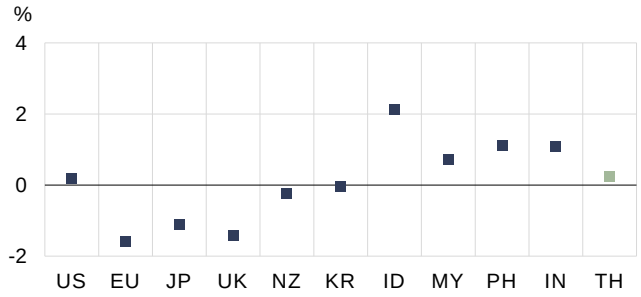
Note: *Business credit covers lending activities of Other Depository Corporation (ODCs)

Sources: Thai Bond Market Association (Thai BMA) and Bank of Thailand

Going forward, **financial conditions were expected to remain accommodative**. The real policy interest rate increased slightly but still remained accommodative and was moderate relative to other countries (Chart 2.11). Meanwhile, costs of financing through commercial banks, as reflected in the new loan rate (NLR), would stabilize at a low level. However, financial institutions were expected to tighten credit standards in the first quarter of 2019[7/], particularly for loans extended to large corporates owing to the outlook of economic slowdown and concerns over some industries. Meanwhile, credit standards for loans extended to households were also expected to be slightly tightened, particularly for auto and mortgage loans.

Chart 2.11 Thailand's real policy rate* slightly increased but remained accommodative overall. The rate was moderate compared with other countries.

Real policy rates*



Note: *Calculated from policy rate subtracted by one-year-ahead inflation expectation according to a survey by Consensus Economics (data as of 11 March 2019)

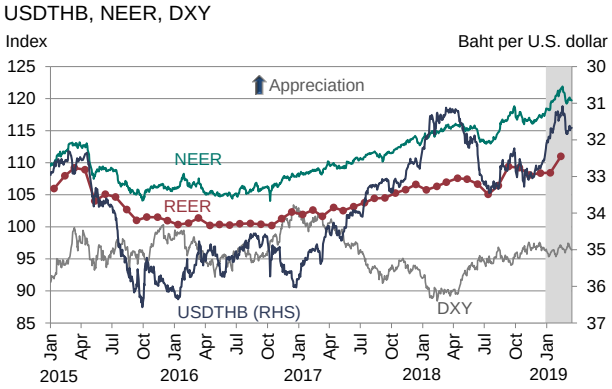
Sources: Bloomberg, Consensus Economics, calculation by Bank of Thailand

[7/] Survey of credit conditions for the fourth quarter of 2018 and outlook for the first quarter of 2019.

The baht appreciated against the U.S. dollar and the nominal effective exchange rate (NEER) index also appreciated.

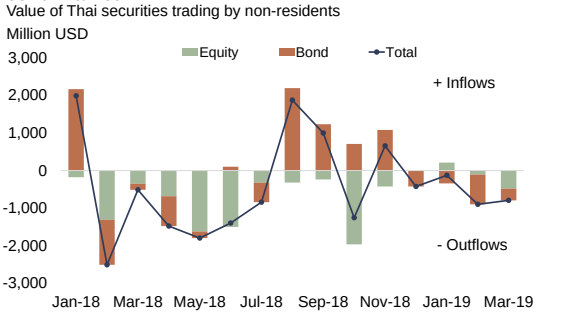
In the first quarter of 2019, **the baht appreciated against the U.S. dollar** relative to the end of the previous quarter (Chart 2.12). In January, the baht appreciated against the U.S. dollar in line with the movement of regional currencies due to the weakening U.S. dollar following the Fed's monetary policy communications toward more dovish stance. Concerns over global economic uncertainties surrounding the no-deal Brexit also alleviated in the first quarter of 2019. In addition, trade negotiations between the U.S. and China showed signs of positive developments. Despite net portfolio outflows of non-residents, primarily from the bond market (Chart 2.13), the baht appreciation was more pronounced than regional currencies, partly due to Thailand's large current account surplus. Also, the flash crash of U.S. securities prices in the beginning of January 2019 led to the appreciation of the yen and the baht which were regarded as safe-haven currencies. Since the end of February 2019, the baht weakened against the U.S. dollar due to better-than-expected outturns of major U.S. economic figures in the fourth quarter of 2018. Furthermore, there were concerns over geopolitical risks regarding an abrupt end of the U.S.-North Korea summit as well as the Thai general elections. Consequently on 19 March 2019, the baht closed at 31.67 to the U.S. dollar, up 2.8 percent from the end of the previous quarter.

Chart 2.12 The baht appreciated against the U.S. dollar due to the weakening U.S. dollar



Source: Bank of Thailand and Reuters (data as of 19 March 2019)

Chart 2.13 Portfolio investment by non-residents recorded net outflows in the first quarter of 2019, primarily from the bond market

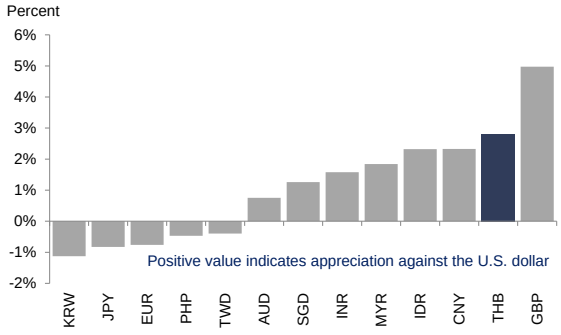


Note: Data on equity flows as of 18 March 2019, data on bond flows as of 14 March 2019

Source: Bank of Thailand

The nominal effective exchange rate (NEER) index stood at 119.72 on 19 March 2019, an appreciation of 2.1 percent from the end of the previous quarter. The movement was in line with the baht appreciation against currencies of most trading partners, except pound sterling (Chart 2.14). As the end of February 2019, the preliminary **real effective exchange rate (REER) index** rose about 2.4 percent from the end of the previous quarter.

Chart 2.14 The baht appreciated against the U.S. dollar in line with the movement of most regional currencies (19 Mar 19 compared to the end of Dec 18)



Source: Bank of Thailand and Reuters (data as of 19 March 2019)

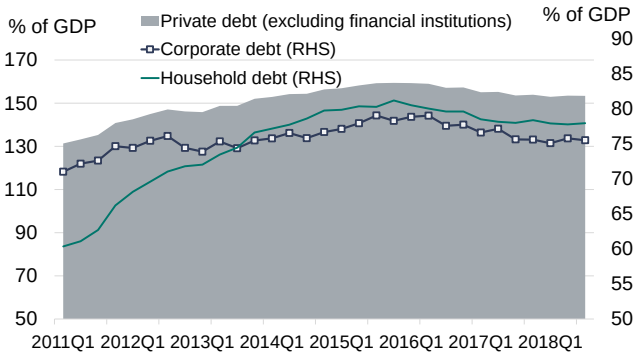
Going forward, exchange rates would likely remain volatile due to monetary policy directions of advanced economies, uncertainties over the U.S. and China trade protectionist measures, geopolitical risks, and the political situation in Thailand. Thus, businesses should regularly manage their foreign exchange risks in both directions through various instruments such as local currency invoicing, foreign currency deposit accounts, and the foreign exchange forwards. (Box: *Building an ecosystem to foster resilience against exchange rate volatilities*)

Financial stability remained sound overall. However, there remained a need to monitor risks that might pose vulnerabilities to the financial system in the future. These included: (1) leveraged household debt and deteriorating credit quality of some categories, (2) search-for-yield behavior amid persistently low interest rates which could lead to underpricing of risks, particularly saving cooperatives and large corporates, and (3) risks in the real estate sector.

Thailand's financial stability remained sound overall. External stability continued to be strong, as indicated by the country's high level of international reserves and sustained current account surpluses, while the external debt to GDP ratio remained low.[8/] These factors combined to cushion the Thai economy against volatilities in global financial markets. Financial institutions maintained robust financial positions, as reflected in high levels of capital buffers to cushion against risks stemming from deterioration in credit quality. Nevertheless, there remained pockets of risks that warranted monitoring going forward. Such risks were as follows.

Chart 2.15 The ratio of household debt to GDP exhibited an increasing sign

The ratio of private credit to GDP



Source: Bank of Thailand

(1) Leveraged household debt would continue, while debt serviceability of some household groups and small enterprises continued to deteriorate. The ratio of household debt to GDP rose slightly from the previous quarter to 77.9 percent in the third quarter of 2018 (Chart 2.15) and remained elevated compared with other emerging economies.[9/] Sustained accumulation of household debt was a structural problem that should be urgently recognized and resolved. (*Box: Financial disciplines of Thai households and the BOT's role in mitigating the household debt problem*). In particular, household debt would continue to rise due to the following factors: (1) a continued expansion of auto loans since the second quarter of 2017 in tandem with increased car sales, partly due to sales promotions and more lenient credit standards on auto loans in the previous periods, and (2) the uptrend in mortgage loans before the macroprudential measure on mortgage loans were to be effective. Besides, persistently low interest rates induced households to incur new debts, which could weigh on future consumption and debt serviceability and result in limited households' cushion against economic volatilities.

Overall credit quality remained stable, partly as a result of debt restructuring and delinquent debt write-offs, as indicated by the NPL ratio of commercial banks in the fourth

[8/] The latest data as of Q4 2018 was 35.3 percent.

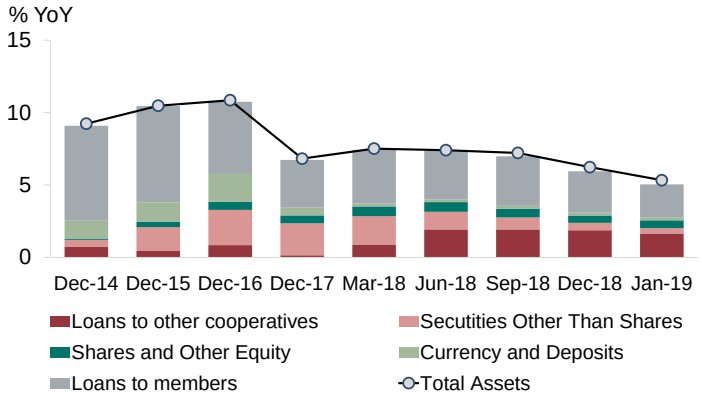
[9/] The ratio of household debt to GDP was around 40 percent among emerging economies in 2018.

quarter of 2018 which was stable at 2.7 percent. However, the NPL ratio of mortgage loans remained high at 3.3 percent. Meanwhile, NPL ratios of auto loans and credit card loans started to trend up since the second half of 2018, which indicated that credit quality had yet to show clear signs of improvement. Financial positions of some small enterprises remained fragile, particularly those in the construction and utilities sectors. These were reflected in the sustained negative operating profit margins (OPM) and interest coverage ratio (ICR) of small companies[10/] which were consistent with the NPL ratio of commercial banks' SME loan portfolios which remained high at 4.5 percent. [11/]

(2) The continued search-for-yield behavior under the environment of persistently low interest rates could lead to underpricing of risks. Although systemic risk remained limited, there were issues that warranted close monitoring. Such risks included, first, a search-for-higher-yield behavior of saving cooperatives. As a result, assets and deposits of saving cooperatives continued to expand at a high rate (Chart 2.16) despite slowing somewhat after regulatory authorities collaborated to enhance supervision standards. Moreover, credit risk and liquidity risk warranted close monitoring owing to higher short-term borrowings by some large saving cooperatives for investment in financial instruments. In addition, increased borrowings among saving cooperatives heightened their interconnection, resulting in increased systemic risks in the saving cooperatives sector. Second, large corporates having significant connectedness with the financial system increasingly raised funds through bond and equity markets given a prolonged period of low interest rates. Those corporates with high debts were in the service, manufacturing, and utilities sectors. Furthermore, many of these corporates expanded foreign investments, particularly holding companies. The business structures of such businesses had become increasingly complicated, resulting in more difficult risk assessment and possibly underpricing of risks. Third, although offshore investments through foreign investment funds (FIF) had contracted somewhat in the recent period due to concerns over volatilities in global financial markets, FIF investments started to rise in February 2019 relative to the same period last year. Concentration risks still warranted monitoring as FIF investments were concentrated in only five major countries.

Chart 2.16 Assets of saving cooperatives expanded at a slower rate, but there remained pressure to search for higher yield

Contribution to growth of savings cooperatives' assets



Note: Saving cooperatives were subjected to tighter regulation by government since H2/2017

Source: Cooperative Auditing Department, calculations by Bank of Thailand

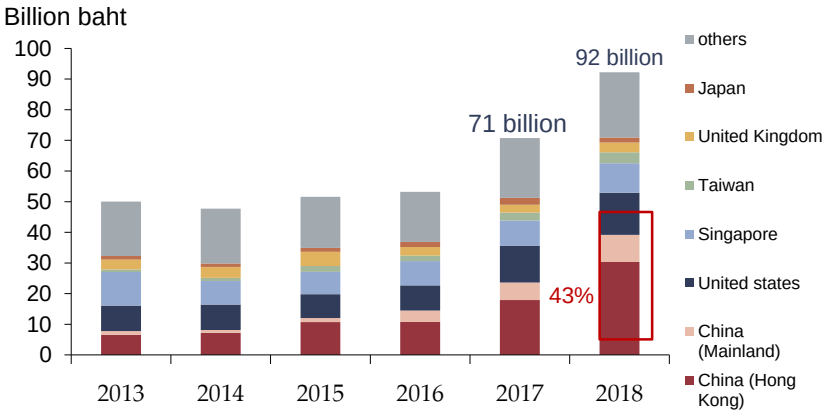
[10/] As reflected by the ICR and OPM of 25th-percentile small-sized listed companies which have been in the negative territory.

[11/] The average ratio of NPL for SMEs during 2013-present was 3.9 percent.

(3) Risks in the real estate sector such as risk pertaining to foreign, particularly Chinese, demand for condominiums. In 2018, foreign demand for condominiums in Thailand continued to grow, as reflected in the value of foreign funds transferred for condominium purchase amounting to 92 billion baht, up 30.3 percent from the same period last year. The majority of these foreign buyers were Chinese, accounting for 43 percent of total foreign funds transferred (Chart 2.17). Consequently, risks regarding possible decline in Chinese demand warranted monitoring as it could result in an oversupply and liquidity problems of real estate developers. In particular, the U.S.-China trade protectionist measures could lead to an economic slowdown in China, and political uncertainties in Thailand could affect decision of Chinese buyers. Furthermore, real estate developers viewed that a decline in Chinese demand would affect tourist destinations such as Chiangmai and Phuket at a greater degree than Bangkok and its vicinities as it would be more difficult to find substitution for Chinese demand in such area. Besides, **there remained a need to monitor the impact of the revised macroprudential measure on mortgage loans**[12/] regarding the adjustment of the property market and mortgage loan quality. The regulations were to be effective on 1 April 2019.

Chart 2.17 The value of funds transferred for condominium purchases by non-residents in terms of country of residence or nationality of the account owner expanded 30.3 percent in 2018 from the previous year

The value of funds transferred for condominium purchases by non-residents



Note: The value of funds transferred for condominium purchases by non-residents is estimated from (1) the amount of foreign currencies sold for down payments or purchases of condominiums and (2) the amount of funds withdrawn from non-residents baht-denominated accounts for condominium purchases.

Source: Bank of Thailand.

12/ See details at https://www.bot.or.th/Thai/PressandSpeeches/Press/News2561/n7261t.pdf

Implications of low unemployment rate in Thailand

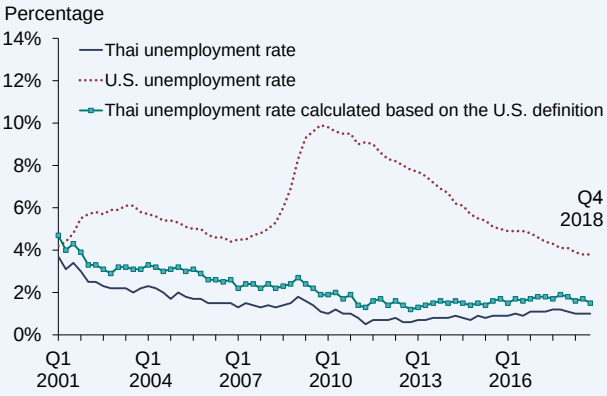
A ranking by the World Bank in 2018 reveals that Thailand's unemployment rate of 1.1% was the 9th lowest out of 233 countries worldwide. Such low unemployment rate may reflect the tightness of the Thai labor market following a solid economic growth in recent periods. However, it may not reveal structural problems in the Thai labor market. **Hence, this article aims to analyze implications of Thailand's low unemployment rate in 3 main aspects, namely 1) definitions of unemployment rate in Thailand and the international standard, 2) structural problems in the Thai labor market that are not reflected in unemployment figures, and 3) alternative indicators to assess the Thai labor market in a wider scope.**

1. Definitions of unemployment rate in Thailand and the international standard

The International Labour Organization (ILO) defines **the unemployed as all persons who are without work or work less than one hour per week.** This definition is adopted by countries all over the world including Thailand. However, the U.S. has added to the definition to include those who work for family businesses without any compensation and work for less than 15 hours per week. Using the U.S. definition, **Thailand's unemployment rate will edge up from 1.1 percent to 1.5 percent, which is still considered very low compared to 4 percent of the U.S.** (Chart 1).

Chart 1 Thailand's unemployment rate is still much lower than that of the U.S., despite using the same calculation based on the U.S. definition

Unemployment rate (seasonally adjusted)



Source: National Statistical Office of Thailand, Calculation by the Bank of Thailand

2. Structural problems in the Thai labor market

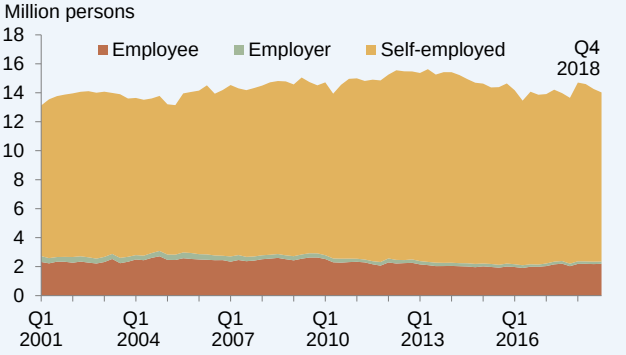
The low unemployment rate in Thailand cannot reflect the following 4 structural problems in the Thai labor market.

<u>First problem</u> Some workers have unstable jobs.

In terms of work status, **most workers in the agricultural sector, which constitutes nearly one-third of the labor force, are self-employed and are not in the social security system.** Thus, they may not have access to significant social benefits for, such as childbirth, childcare, disabilities, death, or unemployment (Chart 2). For workers in the non-agricultural sector, one-third of non-agricultural workers are not in the social security system. This is particularly the case for workers in the trading sector, where most of the firms are small firms that face high competition and are financially vulnerable. Examples of these firms include retail businesses, street stalls, mom-and-pop shops, or stalls in front of shops.

Chart 2 Most workers in the agricultural sector are self-employed and are not in the social security system

Workers in agricultural sector by work status (seasonally adjusted)



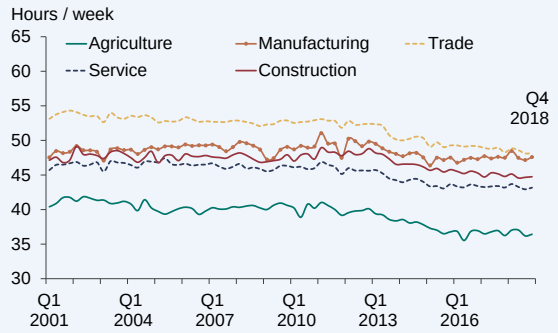
Source: National Statistical Office of Thailand, Calculation by the Bank of Thailand

<u>Second problem</u> Some workers in the agricultural sector are underemployed[13/].

The average working hours of the agricultural sector is 36 hours per week, which is lower than the average of non-agricultural sector at 46 hours per week, and averages of each non-agricultural sector (Chart 3). Furthermore, the number of workers in the agricultural sector who work less than 35 hours per week and prefer to work more is larger than that of the non-agricultural sector. Taken together, a larger number of agricultural workers, compared to non-agricultural workers, is underemployed. Therefore, difficulties for agricultural workers to relocate to other economic activities reflect a structural problem in the Thai labor market.

Chart 3 Workers in the agricultural sector have lower working hours than those in the non-agricultural sector

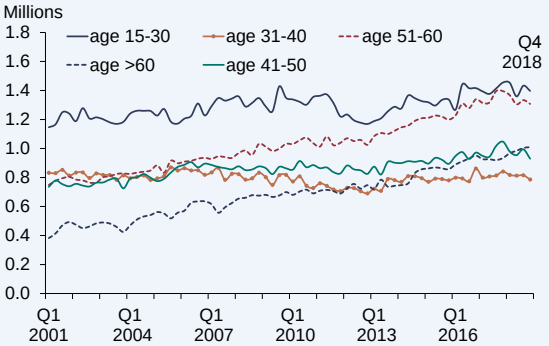
Average working hours per week in agricultural sector (seasonally adjusted)



Source: National Statistical Office of Thailand, Calculation by the Bank of Thailand

Chart 4 The number of houseworkers out of labor force, age between 51 and 60, is trending upward

Number of houseworkers out of labor force (seasonally adjusted)



Source: National Statistical Office of Thailand, Calculation by the Bank of Thailand

<u>Third problem</u> Some workers have given up seeking jobs and are therefore excluded from the labor force.

The low unemployment rate is partly due to **the early retirement of some workers who are thus not counted in the labor force.** This can be observed in the rising number of houseworkers out of the labor force with age between 51 and 60 (Chart 4). However, it is likely that **some of these people have attempted to seek a new job for some time before giving up and becoming discouraged workers**, which are neither considered as unemployed nor part of the labor force. Unlike surveys in developed countries, Thailand's labor force survey is still unable to separate discouraged workers from those out of the labor force. Thus, Thailand's unemployment rate calculated based on this data is relatively lower than if discouraged workers were to be categorized as unemployed and part of the labor force.

<u>Fourth problem</u> Some workers confront skill mismatch given their educational levels or fields of studies.

Although workers are employed, if their jobs do not match their qualifications, they might not be able to reach their full potential. A study reveals that 1 in 10 of non-agricultural workers in Thailand is underpaid given his or her educational level, since the job does not match his or her skills. In other words, worker experiences a job mismatch with regard to both educational levels (vertical mismatch) and fields of study (horizontal mismatch). Among various fields of study, job mismatch is found to be most common among social science graduates.

3. Alternative indicators to assess the Thai labor market in a wider scope

Given four structural problems above, it is clear that **the unemployment rate alone is not sufficient to capture all aspects of labor market development**. Thus, other indicators are

[13/] Underemployed workers refer to workers who work less than 35 hours per week and prefer to work more.

necessary to monitor additional aspects of the overall Thai labor market, including the following 3 dimensions.

1) Labor market confidence, such as business employment outlook which represents confidence among employers

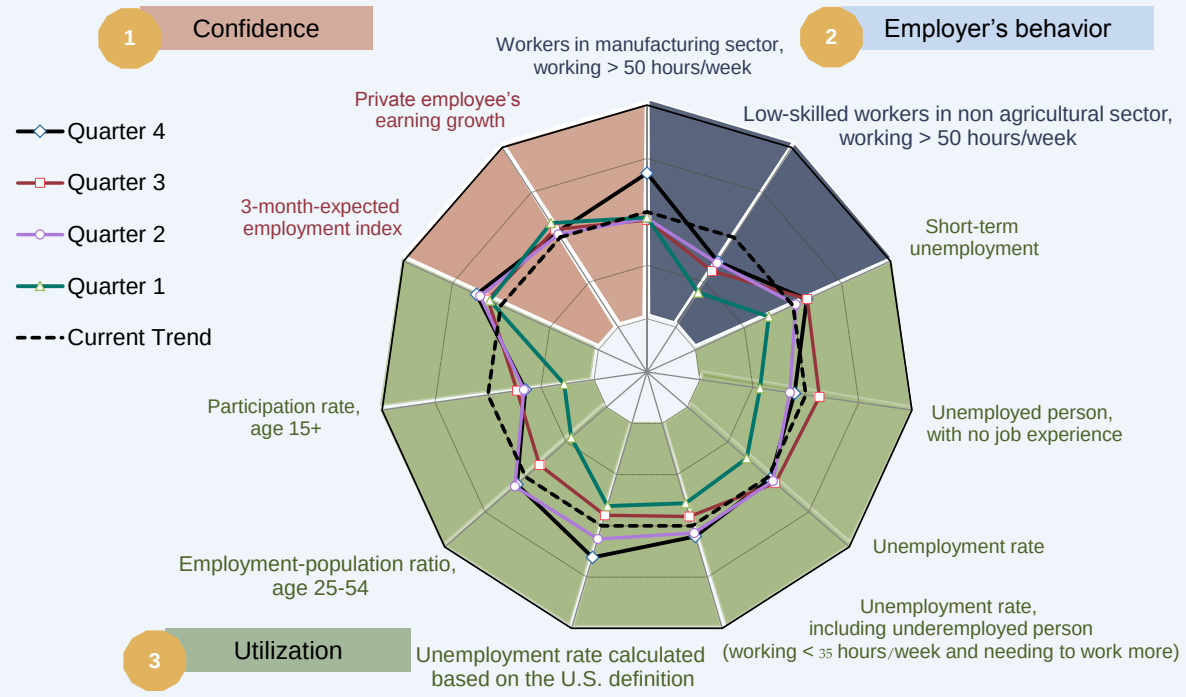
2) Employers' behavior, such as the number of workers who work more than 50 hours per week. More workers working overtime imply a higher demand for labor and signals tightness in the labor market.

3) Labor market utilization, such as the number of unemployed workers without job experiences. This reflects whether the labor force is fully utilized.

Chart 5 shows labor market cyclical indicators, which can be used to assess the overall Thai labor market in wider dimensions . For example, the Thai labor market in the fourth quarter of 2018 improved in many dimensions. Labor market utilization was close to a historical average, while employment was expected to increase further thanks to confidence in economic growth outlook. Moreover, employment for overtime work rose in the manufacturing sector in order to meet higher demand. Nevertheless, demand for low-skill workers continued to fall.

In summary, Thailand's unemployment rate is calculated in accordance with the international standard definition. However, unemployment rate that remains low does not reflect structural problems that plague the Thai labor market. For example, some workers do not work to their full potential, while others have given up seeking jobs and are thus excluded from the labor force. Hence, monitoring various dimensions of labor market development through labor market cyclical indicators will allow for a more comprehensive understanding of the market, and thus economic policy formations will be based on a more comprehensive data set.

Chart 5 Labour market cyclical indicators in 2018



Note: Based on Federal Reserve Bank of Atlanta's framework; the weighted average (current trend) represents mean value (Z-score=0); the more indices deviate from the center point, the more tighthening labour market becomes (the index in each dimension is determined by Z-score, ranking from -2 to 2)

Source: National Statistical Office of Thailand, Calculation by the Bank of Thailand

Building an ecosystem to foster resilience against exchange rate volatilities

Amidst growing uncertainties and volatilities in the global financial market today, the Bank of Thailand (BOT) recognizes the importance of building a financial ecosystem that could foster resilience against volatilities of exchange rates and capital flows. The objective is to ensure that Thai firms are able to manage exchange rate risks appropriately and on a regular basis. This can be done through the following ways.

1. Local currency invoicing This reduces risks from volatilities of major currencies. The BOT encourages firms the practice of invoicing in Thai baht or in currencies of major regional trading partners, as alternative currencies apart from U.S. dollar. This will help reduce risks from reliance on U.S. dollar, which has been highly volatile in recent periods. In 2018, as much as 76.9 percent of Thai export value was invoiced in U.S. dollar (Chart 1), although the actual trade with the U.S. only accounted for 10 percent.

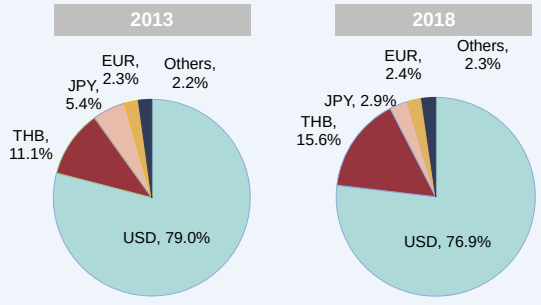
Therefore, **the BOT has been promoting the use of local currencies, starting from Thailand's key regional trading partners**, namely China and Japan. In collaboration with central banks of China and Japan, the BOT has promoted the use of direct quotations between baht-yuan and baht-yen among commercial banks in respective countries. This is in order to increase liquidity of local currencies in foreign exchange markets, and reduce spread costs. Furthermore, the BOT has now extended collaborations to other countries in the region, such as Malaysia, Indonesia, and CLMV countries.

The collaborations above have helped lower costs of exchanging local currencies, prompting firms a greater use of local currency. As a result, the share of Thai exporters that used Thai baht invoicing has grown from 11.1 percent in 2013 to 15.6 percent in 2018 (Chart 1). This is particularly observed in exports to New Zealand, Australia, South Africa, and Russia (Chart 2), where significant shares of Thai automobile exports were recorded (Chart 3). The use of Thai baht as invoicing currency for automobile exports is higher than other export products (Chart 4), partly due to exchange rate risk management policy of their parent companies. Moreover, the share of Thai baht invoicing is high among Thailand's exports to ASEAN countries, especially Myanmar and Laos. Since Thailand is the Chair of ASEAN for 2019, the BOT will take this opportunity to promote a wider use of local currencies. Therefore, all ASEAN members will recognize benefits, and support the use of local currencies for regional trade and investement. This will ultimately foster a greater financial interconnectedness in the region.

Chart 1 The use of Thai baht as invoicing currency has increased

Share of invoicing currency classified by currency

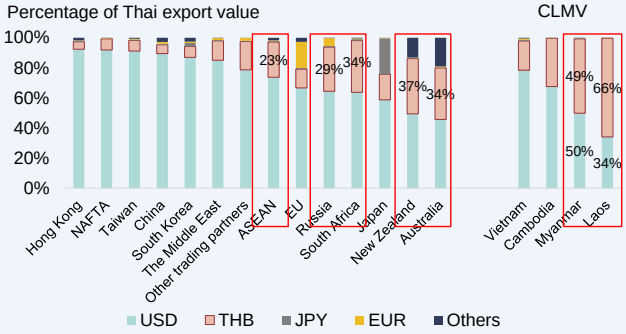
Percentage of Thai export value



Source: Thai Customs Department, calculation by the Bank of Thailand

Chart 2 The use of Thai baht as invoicing currency is significant for Thailand's exports to New Zealand, Australia, South Africa, Russia, and ASEAN especially Laos and Myanmar

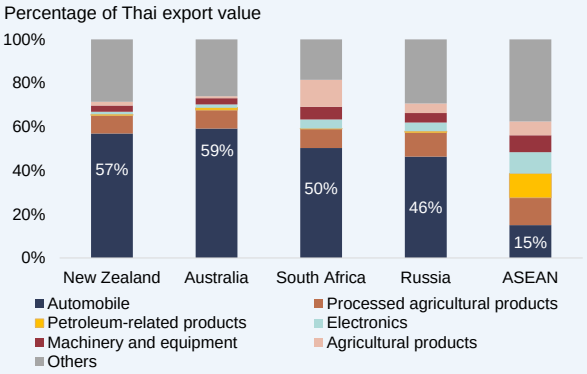
Share of invoicing currency classified by trading partner in 2018



Source: Thai Customs Department, calculation by the Bank of Thailand

Chart 3 Countries with relatively high usage of Thai baht as invoicing currency for trading with Thailand are mostly markets for Thailand's automobile exports

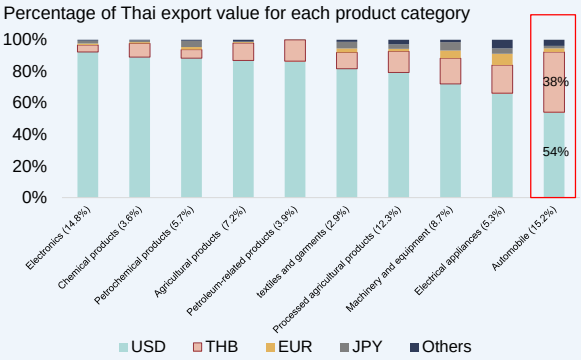
Share of export products classified by trading partner in 2018



Source: Thai Customs Department, calculation by the Bank of Thailand

Chart 4 The use of Thai baht as invoicing currency for automobile exports is higher than other export products

Share of invoicing currencies classified by product category



Note: ( ) denotes share of export value in 2018

Source: Data as of 2018 from Thai Customs Department, calculation by the Bank of Thailand

2. Foreign currency deposits (FCD) Firms with foreign exchange obligations in the future can deposit their foreign currencies in FCD accounts in order to reduce risks from exchange rate volatilities. The BOT has published the deposit rates and transaction fees of FCD accounts of each commercial bank on its website, so that firms can conveniently compare information across banks. According to the latest data, Thai firms have relied increasingly on FCD accounts for the payment of goods and services. Particularly, in 2018, the number of Thai firms that used FCD accounts for payment and receipt of goods and services increased by 17 percent from the previous year.

3. Foreign exchange hedging instruments such as FX forwards and FX options. Recently in 2017, the BOT, in collaboration with government agencies and commercial banks, has launched the "Foreign Exchange Risk Management Program for SMEs" to enhance understandings among SMEs on how to use FX options to mitigate foreign exchange risks. In the second phase of the project, which started in November 2018, the sponsored premium for the purchase of FX options has been raised from 30,000 baht to 50,000 baht per business entity. Furthermore, the BOT has published the average forward points quoted by commercial banks on its website. This will enhance transparency and provide useful information for firms that consider engaging in FX forward contracts with commercial banks. Overall, the effort will facilitate Thai firms by offering a greater variety of foreign exchange hedging instruments.

In addition to supporting Thai firms to be immuned from exchange rate volatitilies, the BOT also recognizes the importance of faciliating more balanced capital flows. In recent periods, sustained current account surplus has led to constant capital inflows into Thailand. Against this backdrop, the BOT has provided greater convenience for residents' outflows, both in the form of outward direct investment and portfolio investment. The objective is to balance between capital inflows and outflows, as well as increase opportunities for Thai investors to enhance returns and diversify risks. To this end**, the BOT has introduced a series of regulatory reform since 2017 according to the "FX Regulations Reform" roadmap**. These are, for example, streamlining procedures and reducing documents required for transferring money abroad, promoting transactions in electronic forms, and removing foreign exchange transaction forms (FX forms). Moreover, firms under specified qualifications can apply as a qualified company (QC), which allows them to conduct foreign exchange transactions with the commercial bank without any underlying documents for more convenience in foreign currency management. Meanwhile, Thai investors with more than 50 million baht in total assets can apply as a qualified investor (QI), which enables them to directly invest in offshore securities without a need to invest through local investment agents, expanding their options in foreign investment.

Financial disciplines of Thai households and the BOT's role in mitigating the household debt problem

Thailand's household debt is a key issue that the Bank of Thailand has placed great emphasis on and monitored constantly in recent years. Currently, household debt totals approximately 78 percent of GDP, which is relatively high compared to the average of only 40 percent among emerging market countries. Furthermore, household debt has appeared to accelerate once again since the second half of 2017. Against this backdrop, this article summarizes the debt behaviors of Thai households based on two recent studies. The first one is a study by the Puey Ungphakorn Institute for Economic Research that finds that **"Thai people are indebted at an earlier age, remaining so for longer, and in a larger amount than before."** The second study is part of a research project on Thailand's household debt problem and the policy implications. It finds that **"financial discipline is one of the crucial factors behind Thailand's household debt problem"** . The policy implications from the study are that collaborations from all parties are necessary to solve such problem, and that government policy will be ineffective should households do not recognize the importance of financial disciplines.

Thai people are indebted from an earlier age, remaining so for longer, and in a larger amount than before.

A study by the Puey Ungphakorn Institute for Economic Research[14/] reveals the following findings based on an analysis of the National Credit Bureau data.

(1) Thai people are indebted from an earlier age and have delinquent loans when they are juvenile. More than half of the working population aged 25-35 **years old** incur **consumer or credit card debts**. Consequently**, this age group has the highest** proportion **of indebted individuals**. **In terms of debt quality,** one-fifth of **borrowers from this age group have** delinquent loans.

(2) Thai people are remaining indebted for longer periods. The size of debt per person tends to surge from the age of late 20s to 30 years old, and remains elevated throughout the individual's working time. Moreover, the debt burden does not seem to fall when the individual reaches retirement, suggesting the lack of financial security among Thais.

(3) Thai people incur a larger amount of debts than before. The median size of debt per person has more than doubled within six years, from 70,000 baht per person in 2010 to around 150,000 baht per person at the end of 2016.

Furthermore, over 16 percent of indebted individuals, or around three million people, are more than 90 days behind on their debt repayments. These are considered delinquent loans, and are therefore subject to debt collections or in the legal process. Such a large share of delinquent loans reflects financial vulnerabilities among Thai people, and suggests that a significant part of the population is trapped in a cycle of indebtedness that resulted in part from the lack of financial discipline and financial literacy.

[14/] Sommarat Chantarat et al. (2017) "Thailand's household debt through the lens of credit bureau data: debt and delinquency" aBRIDGEd No. 10/2017 Puey Ungphakorn Institute for Economic Research.

Financial discipline is one of the crucial factors behind Thailand's household debt problem.

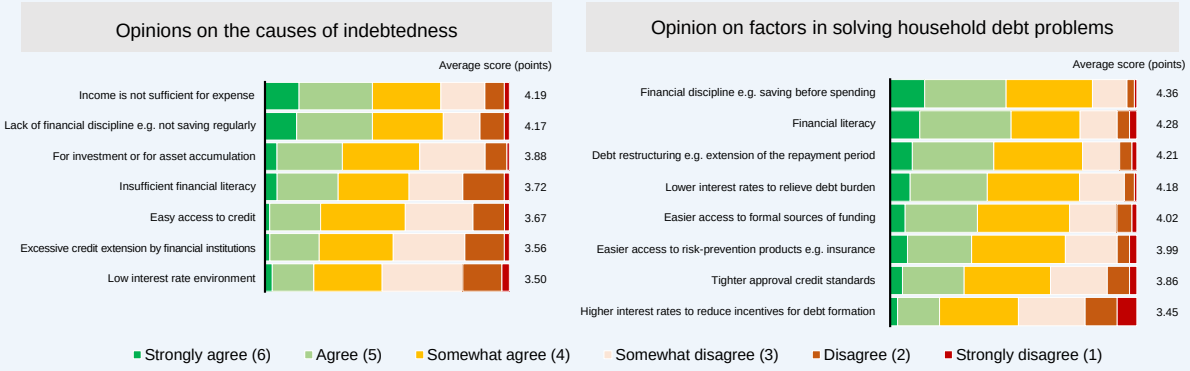
Part of the research project on Thailand's household debt problem and policy implications[15/] is aimed at understanding the perspectives towards debt formation and the spending behaviors among Thai households. A survey conducted on 1,500 households across the country reveals the following insights.

(1) Most Thai households perceive that "financial discipline" plays a key role in both the causes and the solutions of indebtedness. Households become indebted mainly because their income falls short of their expenditures, which linked directly to the lack of financial discipline, especially the absence of consistent saving. At the same time, most households realize that financial discipline, financial literacy, and debt restructuring would help alleviate household debt problem (Chart 1).

Chart 1 Most of Thai households perceived that financial discipline plays a crucial role in both the causes and the solutions of indebtedness

Thai households' perspective on the causes and the solutions of indebtedness

Proportion of households (%) classified by a degree of agreement in each category



Source: Survey of Thai household debt problems and policy implications, Bank of Thailand.

(2) Indebted households tend to be less cautious in spending than those without debts[16/] Compared to other households, indebted ones spend at a larger amount overall, particularly on travel expenses, telephone services, and utilities. In addition, they tend to purchase assets reflecting social status such as housing and automobiles, compared to households without debts.

(3) Indebted households with financial problems often lack financial planning[16/]**.** Indebted households with financial problems are three to five times more likely to engage in overspending behaviors on items such as clothing and recreations, compared to indebted households without financial problems. Moreover, illnesses of family members as reflecting in high healthcare expenditures are among the major factors causing households' liquidity problems. Hence, financial planning and saving for emergency are both crucial and necessary for households' financial security.

[15/] Sra Chuenchoksan et al. (2019) "Thailand's household debt: insights from the BOT-Nielsen Household Financial Survey" FAQ Issue 143 Bank of Thailand.

[16/] The study includes econometrics models. The results represent the difference in average spending among households, controlling for explanatory factors, such as income, assets, occupations, locations of residence, and the number of household members (conditional mean difference).

The BOT's roles in mitigating the household debt problem

The BOT recognizes the importance of the household debt problem, a structural issue that needs to be addressed urgently. The economy driven by indebtedness, especially consumer debts, will be unsustainable, and lead to a build-up of vulnerabilities in the long run. Furthermore, higher debt burden weigh on households' purchasing power, preventing the economy from expanding at its potential. The two conclusions from the first part of this article suggest that household debt problem cannot be tackled only by debt reduction, debt extension, or the extension of the repayment period suspension, but also the adjustments in the spending behaviours and financial discipline among households. To drive such adjustments, the BOT has implemented measures in the following three dimensions.

First dimension: Banking supervision emphasizes responsible lending among financial institutions. Financial institutions must not encourage over-borrowing among borrowers, as irresponsible credit extension will ultimately lead to higher delinquent loans. To enhance responsible lending, the BOT has tightened rules governing credit card loans, personal loans, mortgage loans, and car-for-cash loans. Furthermore, the supervision of financial products marketing has been upgraded such that banks are required to disclose adequate and accurate information regarding their financial products. In addition, the interest rates charged must not be at an unusual high level to ensure that clients are treated fairly and their debt serviceabilities are not affected.

Second dimension: Resolving the problem of debt overhang through Debt Clinic. Debt Clinic is a collaboration between the Bank of Thailand, The Thai Bankers' Association, The Association of International Banks, and the Sukhumvit Asset Management Co. Ltd. (SAM). The objective of the Debt Clinic is to seek a solution for reliable borrowers who determine to change their habits to ultimately get out of their debt cycles. The first phase of the program was emphasized on credit card and uncollateralized personal loans. With SAM serving as an intermediary between debtors and creditors, the Debt Clinic offers multi-creditor debt restructuring program as a one-stop-service. Based on the latest information as of December 2018, the Debt Clinic has completed 1,067 cases of debt restructuring. Set to begin in the second quarter of 2019, the scheme's second phase will allow debtors of credit card and personal loans from non-bank companies to access such program.

Third dimension: Continuously promoting financial literacy and financial discipline. Financial literacy and financial discipline can encourage customers to avoid overspending, design effective financial plan, as well as alert to the fast-changing landscape of financial products. Recently in 2018, the targeted consumers were the generation-Y, such as vocational students and first jobbers. In order to prepare them for future challenges, the BOT aimed to cultivate prudent financial behaviors as well as foster financial immunity and skill. This project was thanks to supports from the Thai Bankers' Association, The Office of the Vocational Education Commission, the Foundation of Virtuous Youth, and other partner organizations.

To this end, although the government has introduced several measures to curb household debt, should households still lack of financial discipline which is at the root of the problem, these government measures will be rendered ineffective. Therefore, tackling the household debt challenges must begin at the household level. Should all Thai individuals are equipped with financial literacy and financial discipline, the financial stability will be strengthened, which will ultimately support stable economic growth going forward.

2.2 Outlook for the Thai Economy

Under the Committee's assessment, the Thai economy was projected to record a slower growth rate at 3.8 percent in 2019, lower than the projection of 4.0 percent in the previous *Monetary Policy Report*. Key growth drivers stemming from external demand, which moderated in line with a slowdown in trading partners' economic growth and global trade volume, caused merchandise and services exports to decelerate from the previous assessment. Meanwhile, **the Thai economy was projected to expand 3.9 percent in 2020,** underpinned by domestic demand as a key driver. The projection for headline inflation in 2019 was close to the previous estimate of 1.0 percent, while that of core inflation in 2019 was slightly below the previous assessment due to lower demand-pull inflationary pressures in response to a softening economic outlook.

Table 2.2 Forecast summary

Percent	2018*	2019	2020
GDP growth	4.1	3.8 (4.0)	3.9
Headline inflation	1.1	1.0 (1.0)	1.1
Core inflation	0.7	0.8 (0.9)	0.9

Note: * Outturn
( ) Monetary Policy Report December 2018
Sources: NESDB, Ministry of Commerce, Bank of Thailand's estimates

Summary of the key forecast assumptions

- Trading partner economies were projected to slow down throughout the forecast horizon due to lower-than-expected growth outturns in the fourth quarter of 2018 and in the first quarter of 2019 in several countries. Exports were undermined by the trade protectionist measures between the U.S. and China, coupled with country-specific factors including the temporary government shutdown in the U.S. and political uncertainties in Europe.
- The federal funds rate, consistent with a more dovish stance of the Fed's monetary policy communications, was projected to be raised only once during the second half of 2019, instead of two times as previously estimated. The Fed was also expected to maintain the policy rate throughout 2020.
- Asian currencies (excluding the Chinese yuan) were expected to appreciate throughout the forecast horizon due to stronger-than-expected outturns during the first quarter of 2019. In the period ahead, regional currencies would gradually and slowly appreciate due to a slower monetary policy normalization of the U.S.
- The Dubai crude oil price projection was maintained. Despite moderating demand for crude oil following the global economic slowdown, tightening oil supply due to a production cut according to the OPEC agreement and Venezuela's lower oil production would support oil prices.
- Farm income was revised upward mainly on account of a higher agricultural output. Meanwhile, overall agricultural prices were expected to contract by a smaller degree, thanks to higher prices of rice, rubber and pork.
- Public spending at current prices were revised downward throughout the forecast horizon for both public consumption and investment. The downward revision was due to delays in infrastructure investment projects of some state-owned enterprises, changes in the investment structure of some PPP projects to allow for greater investment by the private sector, as well as the more-than-expected

budget allocation for replenishment of the treasury account balance for fiscal year 2020, resulting in lower composition of capital and current expenditures.

Table: Summary of forecast assumptions

Annual percentage change	2018*	2019	2020
Dubai crude oil price (U.S. dollar per barrel)	69.6	66.0 (66.0)	66.0
Farm income (% YoY)	0.3	1.0 (-0.4)	0.7
Government consumption at current price (billion baht)[1/]	2,637	2,771 (2,786)	2,918
Public investment at current price (billion baht)[1/]	962	1,034 (1,056)	1,122
Fed funds rate (% at year end)	2.125	2.625 (2.875)	2.625
Trading partners' GDP growth (% YoY)[2/]	3.6	3.2 (3.3)	3.2
Regional currencies (excl. China) vis-à-vis the U.S. dollar (index)3/	153.8	154.5 (158.7)	153.3

Notes: [1/] Assumption includes spending on infrastructure investment plans
2/ This was an assessment as of 19 March 2019 before the Federal Open Market Committee (FOMC) meeting at 20 March
[3/] Weighted by each trading partner's share in Thailand's total exports
[4/] Increasing index represents depreciation, decreasing index represents appreciation
* Outturns
( ) Monetary Policy Report December 2018

Merchandise exports were expected to slow down in line with moderating global economy and global trade volume.

The value of Thailand's merchandise exports in 2019 was projected to slow down. Export volume moderated mainly on account of a softening outlook of trading partner economies and global trade volume. Weaker exports of some industries were expected to continue, including rubber, automobile, integrated circuits, and telecommunications equipment, as reflected by a contraction in their exports in January 2019. Moreover, the World Trade Outlook Indicator[17/] in the first quarter of 2019 displayed a sluggish trend. However, in the period ahead, Thai exports would be supported by the relocation of some hard-disk drive and air-conditioning businesses to Thailand, redirected orders from China to Thailand such as for electronic parts, automobile tyres, as well as televisions, and also by the 5G technology-related infrastructure investment plans in many countries, which would benefit Thailand's exports of electronic parts in the period ahead**. The Committee thus projected the value of merchandise exports in 2019 to grow at 3.0 percent, down from the previous estimate of 3.8 percent in the previous *Monetary Policy Report*. In 2020, the value of merchandise exports was expected to expand at 4.1 percent.**

Exports of services were projected to exhibit slower growth mainly due to lower spending per head.

Overall exports of services were projected to exhibit slower growth mainly due to lower spending per head, despite a higher-than expected number of foreign tourists. The global economic slowdown led to lower spending per head, as observed in lower spending per head by Chinese tourists, particularly on luxurious goods. This was in tandem with Chinese tourists spending in other countries. In addition, the slowdown of the euro area economy resulted in a shorter trip duration of European tourists. However, the projected number of foreign tourist arrivals was revised upward on the back of tourist confidence on improvements in Thailand's

[17/]The World Trade Outlook Indicator (WTOI) is compiled by the World Trade Organization (WTO)

safety standards and benefits from the extended exemption of visa-on-arrival (VOA) fees until the end of April 2019, particularly for tourists from China, India and Taiwan. As a result, the projected number of foreign tourists in 2019 was revised up to 40.4 million from the previous estimate of 40.0 million, while in 2020, the number of foreign tourists was forecasted to be 42.0 million.

The current account was projected to record a 34.5 billion U.S. dollars surplus in 2019, close to the estimated surplus in the previous *Monetary Policy Report*. The estimate was in tandem with a decline in the value of both merchandise exports and imports. Meanwhile, **the projection for the current account surplus in 2020 was 31.5 billion U.S. dollars**, a smaller surplus as import value was expected to accelerate in line with outlook of private investment.

Private consumption was expected to be well maintained.

Private consumption was expected to be well maintained despite some moderation in durable goods, especially automobiles which had accelerated significantly in 2018. Supporting factors included a continued improvement in both farm and non-farm household income, improved consumer confidence, and government policies which would support household consumption such as the welfare card scheme. Nonetheless, elevated household debt would affect households to allocate part of their income for debt repayment, especially among low-income households, and somewhat weigh on private consumption. Moreover, the adoption of automation in place of labor and inflation remaining at a low level would limit pressures on employers to increase wages, thus resulting in a gradual improvement in household purchasing power in the period ahead.

Private invest would continue to expand.

Private investment, particularly investment in machinery and equipment, would continue to expand. Drivers would come from both upgrades to production efficiency and capacity expansion, as well as the relocation of production base to Thailand for hard-disk drive, which was according to the original investment plan and the PPP infrastructure investment projects in the Eastern Economic Corridor (EEC). Moreover, additional relocations of production base to Thailand for many industries affected by the trade protectionist measures between the U.S. and China were expected, which would lend support to Thai exports going forward. In addition, private investment was projected to expand further in 2020 thanks to changes in the investment structure of the public-private partnership for the Orange Line mass rapid transit project between Bangkhunnon and Thailand Cultural Center, which would be undertaken by the private sector.

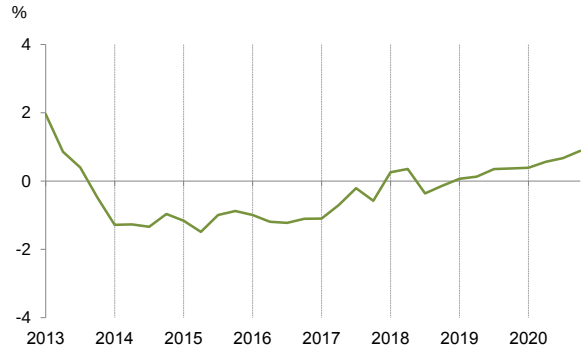
Public spending would continue to drive the economy.

Public spending would continue to drive the economy, especially investment in transportation infrastructure, including the dual-track railway and mass rapid transit projects. Nonetheless, public investment was projected to grow at a slower pace than previously assessed due to delays in some projects. For state-owned enterprises, the high-speed rail project between Bangkok and Nakhornratchasima was postponed as it was during the process of contract reviews, while the Orange Line mass rapid transit project between Bangkhunnon and Thailand Cultural Center underwent changes in the investment structure of the public-private partnership to allow for greater investment by the private sector. For the central government, government investment continued to be affected by a reduction in the construction

budget and extended construction periods of the Bang Yai-Kanchanaburi motorway project due to problems regarding access into construction areas. Moreover, public expenditure was revised downward since the budget allocation for replenishment of the treasury account balance for fiscal year 2020 was higher than expected, resulting in lower composition of capital and current expenditures.

The projection of headline inflation was close to the previous assessment, while core inflation was projected to decline slightly.

Chart 2.18 Output gap

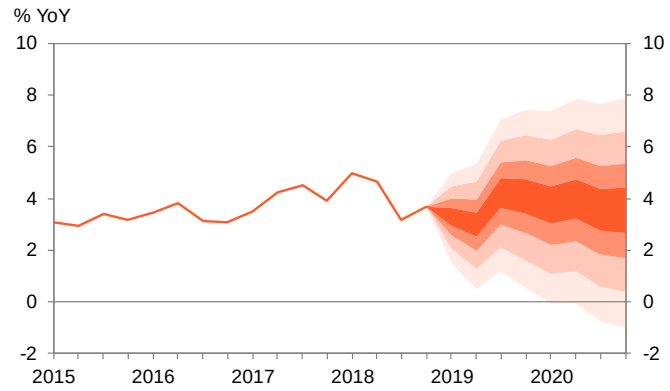


Headline inflation projection was at a level close to the estimate in the previous *Monetary Policy Report.* Energy prices still faced upward pressures due to tightening crude oil supply, which would compensate for a decrease in demand in tandem with the global economic slowdown. Meanwhile, fresh food prices were expected to rise as the drought condition might intensify, the excess supply of pork and eggs alleviated, and prices of unmilled jasmine rice remained elevated. On the other hand, the projection of core inflation was revised downward due to lower-than-expected outturns in January and February 2019 and a softening economic outlook. However, demand-pull inflationary pressures would subsequently slowly rise reflecting a gradual closing of the output gap in the period ahead (Chart 2.18). Thus, **the Committee projected headline inflation to average 1.0 and 1.1 percent for 2019 and 2020, respectively, and core inflation to average 0.8 and 0.9 percent for 2019 and 2020, respectively.**

Growth and inflation projections were still subject to downside risks.

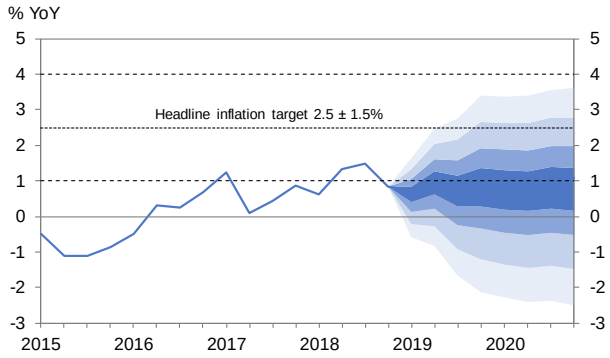
Under the Committee's assessment, **growth projection was subject to downside risks**, as reflected by the fanchart that tilted downward throughout the forecast horizon (Chart 2.19). **Possibilities that the Thai economy would underperform the baseline projection** included (1) lower-than-expected trading partner economic growth following more intensified trade protectionist measures between the U.S. and China, uncertainties regarding Brexit negotiations, the U.S. import tariffs on automobiles and parts, and geopolitical risks; and (2) lower-than-expected private investment due to political uncertainties. However, **there were possibilities that the Thai economy would outperform the baseline projection** due to (1) less-than-expected slowdown of the Chinese economy due to government economic stimulus measures; (2) higher-than-expected domestic demand as a result of (2.1) sooner-than-expected implementation of government infrastructure projects, PPP projects, and private investment, (2.2) a quick dissolving of political uncertainties, and (3) additional government measures to support private spending. Meanwhile, **risks to the forecasts of headline and core inflation were expected to tilt downward** in line with downside risks to growth projections (Chart 2.20 and Chart 2.21).

Chart 2.19 Growth forecast



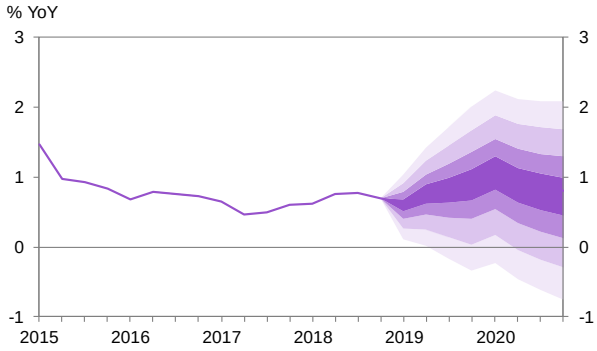
Note: Fan chart covers 90% of the probability distribution

Chart 2.20 Headline inflation forecast



Note: Fan chart covers 90% of the probability distribution

Chart 2.21 Core inflation forecast



Note: Fan chart covers 90% of the probability distribution

Table 2.3 Forecasts of GDP and components

Annual percentage change	2018*	2019	2020
GDP growth	4.1	3.8 (4.0)	3.9
Domestic demand	3.9	3.9 (4.0)	3.9
Private consumption	4.6	3.9 (4.0)	3.7
Private investment	3.9	4.4 (4.5)	5.0
Government consumption	1.8	2.3 (2.6)	2.6
Public investment	3.3	6.1 (6.6)	6.6
Exports of goods and services	4.2	3.1 (4.1)	3.7
imports of goods and services	8.6	2.7 (3.2)	4.2
Current account (billion, U.S. dollars)	35.2**	34.5 (34.5)	31.5
Value of merchandise exports	7.2	3.0 (3.8)	4.1
Value of merchandise imports	14.3	3.1 (3.8)	4.8
Number of foreign tourists (million person)	38.3	40.4 (40.0)	42.0

Note: *Outturns

** Current account as of 29 March 2019

( ) Monetary Policy Report December 2018

3. Monetary Policy Decision

The Committee assessed the Thai economy would expand around its potential on the back of domestic demand despite somewhat at a slower pace due to a slowdown in external demand. Headline inflation was expected to rise in line with the previous projection. There remained pockets of risks in the financial system that might pose vulnerabilities to financial stability in the future.

The Committee weighed various factors in determining the most appropriate course of monetary policy, placing great emphasis on the strength and continuation of economic growth, preserving financial stability, and development of headline inflation. Details on the assessment of latest developments and outlook are as follows.

1. Economic growth The Thai economy was expected to expand around its potential, on the back of domestic demand, despite somewhat at a slower pace than previously assessed due to a slowdown in external demand. Private consumption was expected to continue expanding. Both farm and non-farm income continued to improve, while purchasing power of low-income households were additionally supported by government measures. Private investment would continue to expand thanks to the relocation of production base to Thailand, and a further investment of the private sector in the public-private partnership (PPP) investment projects. Public spending would continue to drive the economy, especially public investment on transport infrastructure including the dual-track railway and mass rapid transit projects despite delays in some projects. However, growth drivers stemming from external demand were expected to moderate following the slowdown in trading partner economies and global trade volume. Merchandise exports would be affected by the trade protectionist measures between the US and China and a down cycle of electronic products. Exports of services would exhibit a slower growth due to a lower spending per head as a result of the global economic slowdown, despite an increasing number of foreign tourists. The Committee assessed that the growth projection was still subject to downside risks due to (1) lower-than-expected growth of trading partner economies following such as more intensified trade protectionist measures between the U.S. and China, uncertainties regarding Brexit negotiation, the U. S. import tariffs on automobiles and parts, and geopolitical risks, and (2) lower-than-expected private investment due to political uncertainties.

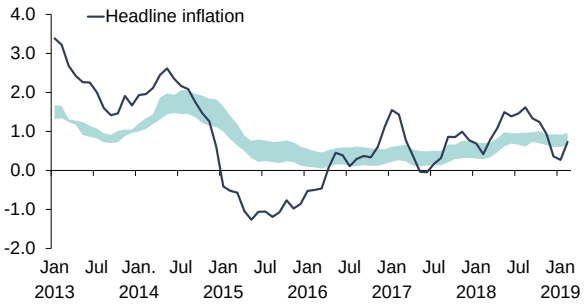
2. Preserving financial stability The Committee viewed that there remained certain pockets of risks in the financial system that might pose vulnerability to financial stability in the future. Such risks included **(1) elevated household debt**, especially auto and personal loans which continued to expand. Meanwhile, credit quality of mortgage and auto-leasing loans would deteriorate due in part to loosen credit standards in the previous periods. Moreover, a prolonged low interest rate environment partly induced households to create new debt, which could weigh on consumption and affect debt-serviceability in the future. **(2) The continued search-for-yield behavior given a prolonged low interest rate environment could lead to underpricing of risks.** For example, saving cooperatives continued to search for higher returns as reflected in high growth of their assets and deposits, although somewhat decelerated. In addition, large corporates would increasingly raise funds given a prolonged low interest rate environment, reflecting their high proportions in the bond market and commercial bank loans. Nevertheless, the business structure of such companies had become increasingly complex and thus resulted

in more complicated risk assessment and possibly underpricing of risks. **(3) There remained risks in the real estate sector**, including oversupply of condominiums in some areas and increasing role of Chinese demand on Thai condominiums which might decrease given a slowdown of the Chinese economy. In addition, there was a need to monitor the adjustments in the property market and mortgage loans after the revised macroprudential measures on mortgage loans were to be effective in April 2019.

3. Inflation development Headline inflation was projected to trend up in line with the previous assessment owing to upward pressure on energy prices as a result of a tightening crude oil supply that offset demand prospects following the global economic slowdown. Fresh food prices would rise on account of an intensified drought. Meanwhile, core inflation was expected to decline following a slower pace of economic growth. However, demand-pull inflationary pressures would gradually rise as reflected in various indicators including trimmed mean and principal component indicators. (Chart 3.1) Nevertheless, structural changes such as intensified price competition and cost-reduction technology partly caused inflation to rise at a slower pace than in the past. Moreover, short-term inflation expectations (1-year ahead) were largely unchanged from the previous quarter. (Chart 3.2)

Chart 3.1 Demand-pull inflationary pressures gradually increased

Percent change from same period last year

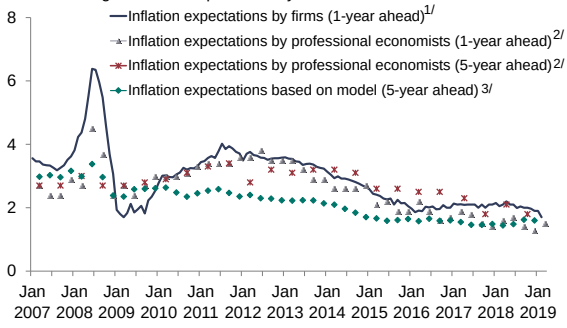


Note: The field shows the highest and lowest outcomes among different measures of underlying inflation. The measures included are 1. **Asymmetric trim** (excludes goods and services with most volatile price changes, removing the bottom 15 percentile and the top 10 percentile), 2. **Principal component model** (calculates changes in common statistical components that attribute price movements across categories of goods and services) and 3. **Core inflation excluding rents and government measures.**

Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

Chart 3.2 Inflation expectations remained stable from the previous periods

Percent change from same period last year



Source: 1/ Business Sentiment Survey of Bank of Thailand (BSI)
2/ Asia Pacific Consensus Forecast
3/ Calculations based on macro-finance term structure model with bond yield and macroeconomic data

The policy rate hike had transmitted to money market rates and short-term government bond yields, including deposit rates.

The Committee assessed the monetary policy transmission after the policy rate hike in December 2018. The money market rates, short-term government bond yields, and deposit rates of some financial institutions began to rise. The current level of banking liquidity was sufficient to support loan growth in the period ahead, while most lending rates were largely unchanged. Real interest rates edged up slightly but still remained at low levels. The private sector continued to raise funds, where both corporate and consumer loans expanded. Moreover, the Committee viewed that a slight reduction in the degree of monetary policy accommodation at the previous meeting would not be an obstacle to economic growth in the period ahead but

would help reduce risks to financial stability, which would in turn benefit the sustainability of economic growth in the long run.

The Committee voted to keep the policy rate at 1.75 percent. Going forward, a policy rate hike would be gradual and follow a data-dependent approach.

At the meeting on February 6, 2019, the Committee voted 4 to 2 to maintain the policy rate at 1.75 percent. Two members voted to raise the policy rate 0.25 percentage point from 1.75 to 2.00 percent. One MPC member was unable to attend the meeting. The Committee viewed that current accommodative monetary policy stance had contributed to the continuation of economic growth and was appropriate given the inflation target. Thus, most members decided to keep the policy rate unchanged at this meeting. Nevertheless, two members viewed that the economy continued expanding around its potential, and that overall financial conditions would remain accommodative and conducive to economic growth despite an additional 0.25 percentage point increase in the policy rate. Hence, they voted to raise the policy rate at this meeting to curb financial stability risks and to build up policy space.

At the meeting on March 20, 2019, the Committee voted unanimously to maintain the policy rate at 1.75 percent. The Committee viewed that the current accommodative monetary policy stance had contributed to the continuation of economic growth and was appropriate given the inflation target. In addition, given heightened global and domestic uncertainties in the current period, the Committee thus voted to keep the policy rate unchanged to assess the clarity of impacts from such uncertainties. Furthermore, there remained a need to address risks to financial stability through a combination of policy tools, including an appropriate policy rate as well as microprudential and macroprudential measures.

Looking ahead, the policy rate hike would be gradual and follow a data-dependent approach. The Committee would closely monitor developments of economic growth, inflation, and financial stability, together with associated risks, in deliberating appropriate monetary policy in the period ahead.

4. Appendix

4.1 Table

Thai Economy Dashboard

Percent	2017	2018	2017	2018			
			Q4	Q1	Q2	Q3	Q4
GDP growth	4.0	4.1	3.9	5.0	4.7	3.2	3.7
Production							
Agriculture	3.7	5.0	-4.5	6.5	10.0	2.7	1.4
Non-agriculture	4.1	4.0	5.0	4.8	4.2	3.2	4.0
Manufacturing	2.9	3.0	3.5	3.8	3.2	1.6	3.3
Construction	-2.8	2.7	-5.9	1.2	1.9	4.5	3.4
Wholesales and retail trade	7.0	7.3	8.2	7.0	7.3	7.3	7.5
Hotels and restaurants	10.6	7.9	16.5	13.1	8.8	4.1	5.3
Transport, storage, and communication	7.3	6.3	8.3	7.1	6.5	5.3	6.1
Financial intermediation	5.4	3.3	3.6	3.6	4.6	3.1	1.8
Real estate, renting, and business activities	4.9	4.0	6.1	4.8	3.2	4.2	3.6
Expenditure							
Domestic demand	2.1	3.9	2.0	3.3	3.6	4.3	4.3
Private consumption	3.0	4.6	3.2	3.8	4.1	5.2	5.3
Private investment	2.9	3.9	4.0	3.1	3.1	3.8	5.5
Government consumption	0.1	1.8	-1.0	1.8	2.3	1.9	1.4
Public investment	-1.2	3.3	-6.0	4.0	4.9	4.2	-0.1
Imports of goods and services	6.2	8.6	7.0	9.1	8.8	11.0	5.6
imports of goods	7.4	8.1	7.3	10.4	7.9	9.9	4.5
imports of services	1.3	10.7	5.7	3.9	12.8	16.1	10.1
Exports of goods and services	5.4	4.2	7.4	8.0	9.6	-0.9	0.6
exports of goods	5.7	4.1	6.6	7.2	9.5	-0.5	0.8
exports of services	4.6	4.4	10.3	9.9	10.3	-2.2	-0.2
Trade balance (billion, U.S. dollars)	34.2	23.6	7.0	8.4	7.4	3.4	4.4
Current account (billion, U.S. dollars)	50.2	37.7	12.4	16.8	8.2	4.2	8.5
Financial account (billion, U.S. dollars)	-12.6	-22.2	-5.5	-4.3	-9.5	-4.2	-4.3
International reserves (billion, U.S. dollars)	202.6	205.6	202.6	215.6	206.8	204.5	205.6
Unemployment rate (%)	1.2	1.1	1.1	1.2	1.1	1.0	0.9
Unemployment rate, seasonally-adjusted (%)	n.a.	n.a.	1.3	1.2	1.0	1.0	1.0

Source: Office of the National Economic and Social Development Board National Statistical Office and Bank of Thailand

Financial Stability Dashboard

Indicators	2017	2018	2017	2018				2019	
			Q4	Q1	Q2	Q3	Q4	Jan	Feb
1. Financial market sector									
Bond market									
Bond spread (10 years - 2 years)	1.1	0.7	1.1	1.2	1.1	0.9	0.7	0.6	0.8
Equity market									
SET index (end of period)	1,753.7	1,563.9	1,753.7	1,776.3	1,595.6	1,756.4	1,563.9	1,641.7	1,653.5
Actual volatility of SET index[1/]	6.5	12.1	7.9	9.4	12.3	11.6	14.0	7.1	8.5
Price to Earnings ratio (P/E ratio) (times)	19.1	14.8	19.1	18.3	16.2	17.3	14.8	15.6	17.1
Exchange rate market									
Actual volatility of Thai baht (%annualized)[2/]	4.4	3.3	2.8	4.6	4.4	4.5	4.7	4.5	4.6
Nominal Effective Exchange Rate (NEER)	110.6	115.6	112.9	114.8	115.2	115.2	117.0	119.1	121.0
Real Effective Exchange Rate (REER)	103.7	107.2	105.7	106.3	107.1	107.1	108.5	109.0	110.4
2. Financial institution sector[3/]									
Minimum Lending Rate (MLR)[4/]	6.28	6.28	6.28	6.28	6.28	6.28	6.28	6.28	6.28
12-month fixed deposit rate[4/]	1.37	1.37	1.37	1.37	1.37	1.37	1.37	1.42	1.42
Capital adequacy									
Capital funds / Risk-weighted asset (%)	18.2	18.3	18.2	18.1	17.9	18.4	18.3	18.3	n.a.
Earning and profitability									
Net profit (billion, Thai baht)	187.1	207.2	40.6	50.2	56.4	51.1	49.5	n.a.	n.a.
Return on assets (ROA) (times)	1.1	1.1	1.1	1.1	1.2	1.2	1.1	n.a.	n.a.
Liquidity									
Loan to Deposit and B/E (%)	96.1	98.2	96.1	95.0	96.8	98.2	98.2	96.9	n.a.
3. Household sector									
Household debt to GDP (%)	78.4	n.a.	78.4	77.9	77.7	77.9	n.a.	n.a.	n.a.
Financial assets to debt (times)	2.6	n.a.	2.6	2.7	2.6	2.5	n.a.	n.a.	n.a.
Non-Performing Loans (NPLs) of commercial banks (%)									
Consumer loans	2.7	2.7	2.7	2.8	2.7	2.7	2.7	n.a.	n.a.
Housing loans	3.2	3.2	3.2	3.4	3.4	3.4	3.2	n.a.	n.a.
Auto leasing	1.6	1.7	1.6	1.5	1.5	1.6	1.7	n.a.	n.a.
Credit cards	2.6	2.3	2.6	3.2	2.4	2.5	2.3	n.a.	n.a.
Other personal loans	2.5	2.5	2.5	2.7	2.5	2.5	2.5	n.a.	n.a.
4. Non-financial corporate sector[5/]									
Operating profit margin (OPM) (%)	8.0	7.5	7.9	7.9	7.8	8.0	6.5	n.a.	n.a.
Debt to Equity ratio (D/E ratio) (times)	0.7	0.7	0.7	0.7	0.7	0.8	0.7	n.a.	n.a.
Interest coverage ratio (ICR) (times)	6.5	6.4	7.7	7.1	6.5	6.6	5.4	n.a.	n.a.
Current ratio (times)	1.7	1.6	1.7	1.7	1.7	1.6	1.6	n.a.	n.a.
Non-Performing Loans (NPLs) of commercial banks (%)									
Large businesses	1.8	1.7	1.8	1.7	1.7	1.5	1.7	n.a.	n.a.
SMEs	4.4	4.5	4.4	4.5	4.5	4.7	4.5	n.a.	n.a.

Note:

[1/] Calculated by 'annualized standard deviation of return' method

[2/] Daily volatility (using exponentially weighted moving average method)

[3/] Based on data of all commercial banks

[4/] Average value of 5 largest Thai commercial banks

[5/] Only listed companies on Stock Exchange of Thailand (median value); with data revisions

Financial Stability Dashboard (continue)

Indicators	2017	2018	2017	2018				2019	
			Q4	Q1	Q2	Q3	Q4	Jan	Feb
5. Real estate sector									
Number of approved mortgages from commercial banks (Bangkok and Vicinity) (units)									
Total	65,124	73,345	18,685	15,011	17,917	18,803	21,614	4,291	n.a.
Single-detached and semi-detached houses	14,517	15,906	3,826	3,527	3,825	3,947	4,607	912	n.a.
Townhouses and commercial buildings	21,469	25,039	5,748	5,166	6,055	6,324	7,494	1,569	n.a.
Condominiums	29,138	32,400	9,111	6,318	8,037	8,532	9,513	1,810	n.a.
Number of new housing units launched for sale (Bangkok and Vicinity) (units)									
Total	110,575	114,477	24,863	26,829	19,836	42,451	35,861	7,866	n.a.
Single-detached and semi-detached houses	19,433	14,280	3,608	4,223	2,790	6,248	5,050	1,055	n.a.
Townhouses and commercial buildings	32,792	36,571	7,892	6,657	6,548	8,759	10,385	1,047	n.a.
Condominiums	58,350	63,626	13,363	15,949	10,498	27,444	20,426	5,764	n.a.
Housing price index (2009 = 100)									
Single-detached houses (including land)	130.9	138.8	133.9	138	138	140	140	141	n.a.
Townhouses (including land)	141.1	149.9	143.7	146	150	152	152	154	n.a.
Condominiums	169.6	180.9	173.0	180	177	180	187	187	n.a.
Land	171.7	175.8	178.3	176	177	173	178	178	n.a.
6. Fiscal sector									
Public debt to GDP (%)	40.7	41.2	41.2	41.2	41.0	42.0	41.9	41.7	n.a.
7. External sector									
Current account balance to GDP (%)[6/]	11.0	7.4	10.1	13.1	6.6	3.4	6.6	n.a.	n.a.
External debt to GDP (%)[7/]	36.7	35.3	36.7	36.6	35.1	35.3	35.3	n.a.	n.a.
External debt (billion, U.S. dollars)	155.2	161.8	155.2	157.9	154.3	158.1	161.8	163.7	n.a.
Short-term (%)	44.3	38.8	44.3	43.1	43.0	41.4	38.8	38.2	n.a.
Long-term (%)	55.7	61.2	55.7	56.9	57.0	58.6	61.2	61.8	n.a.
International reserves / Short-term external debt (times)	2.9	3.3	2.9	3.2	3.1	3.1	3.3	3.4	n.a.

Note:

[6/] Current account / Nominal GDP at the same quarter

[7/] External debt / 3-year average nominal GDP

Table: Probability distribution of GDP growth forecast

Percent	2019				2020			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
> 9	0	0	0	1	1	2	1	2
8-9	0	0	1	2	2	3	2	3
7-8	0	0	4	5	4	6	5	5
6-7	0	2	10	10	9	10	8	8
5-6	4	6	17	15	13	14	12	12
4-5	20	16	22	19	17	16	15	14
3-4	36	25	20	18	17	16	15	15
2-3	28	25	14	14	15	13	14	13
1-2	10	16	7	9	11	10	11	11
0-1	2	7	3	5	6	6	8	8
(-1)-0	0	2	1	2	3	3	5	5
(-2)-(-1)	0	0	0	1	1	1	2	3
(-3)-(-2)	0	0	0	0	0	1	1	1
< (-3)	0	0	0	0	0	0	1	1

Table: Probability distribution of headline inflation forecast

Percent	2019				2020			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
> 4.0	0	0	0	2	2	2	3	3
3.5-4.0	0	0	1	2	2	2	3	3
3.0-3.5	0	1	2	4	4	4	4	4
2.5-3.0	0	3	4	6	6	6	6	6
2.0-2.5	1	9	8	9	8	8	8	8
1.5-2.0	7	15	12	11	10	10	10	10
1.0-1.5	20	19	14	12	11	11	11	11
0.5-1.0	29	19	15	12	11	11	11	11
0.0-0.5	24	15	13	11	11	10	10	10
(-0.5)-0.0	13	10	11	9	9	9	9	9
(-1.0)-(0.5)	5	5	8	7	8	8	7	7
(-1.5)-(1.0)	1	2	5	5	6	6	6	6
(-2.0)-(-1.5)	0	1	3	4	4	4	4	4
< -2.0	0	0	3	6	7	8	7	8

Table: Probability distribution of core inflation forecast

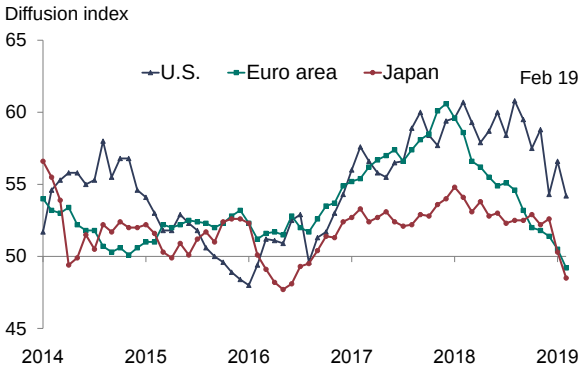
Percent	2019				2020			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
> 4.0	0	0	0	0	0	0	0	0
3.5-4.0	0	0	0	0	0	0	0	0
3.0-3.5	0	0	0	0	0	0	0	0
2.5-3.0	0	0	0	1	2	1	1	1
2.0-2.5	0	0	1	4	7	5	5	5
1.5-2.0	0	3	9	14	18	14	13	12
1.0-1.5	7	25	27	25	26	23	21	20
0.5-1.0	56	44	33	27	23	24	24	23
0.0-0.5	35	23	21	18	15	18	19	19
(-0.5)-0.0	2	4	7	8	6	9	11	12
(-1.0)-(0.5)	0	0	1	3	2	3	5	6
(-1.5)-(1.0)	0	0	0	1	0	1	1	2
(-2.0)-(-1.5)	0	0	0	0	0	0	0	1
< -2.0	0	0	0	0	0	0	0	0

4.2 Chart pack

Global Economy

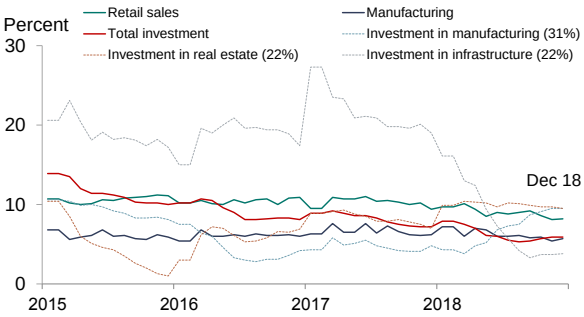
Major advanced economies were expected to expand at a slower rate. The U.S. economy slowed down due to a temporary impact from the government shutdown. The euro area and Japanese economies moderated on account of domestic factors and a slowdown of global trade volume. The Chinese economy decelerated as a result of a series of financial stability measures implemented in recent periods. Meanwhile, slowing global trade had a larger-than-expected impact on Asian economies. Looking ahead, Thailand's trading partners were projected to expand with private consumption and public spending as key drivers.

Manufacturing Purchasing Manager Index



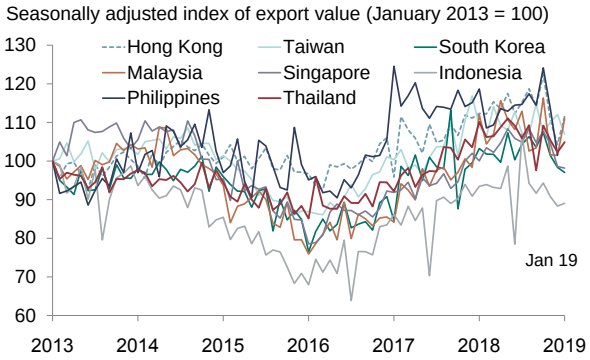
Sources: Bloomberg and Eurostat

China's economic indicators (Change from the same period last year)



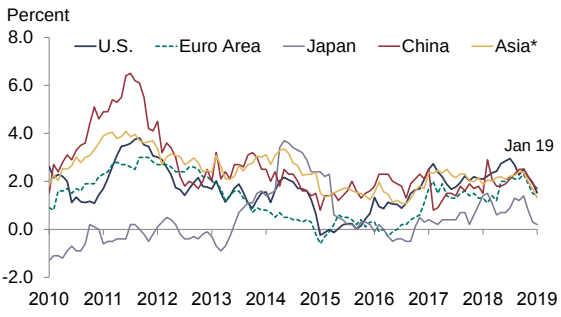
Note: ( ) denotes share to total investment
Source: CEIC

Asian exports



Source: CEIC

Inflation of Thailand's major trading partners

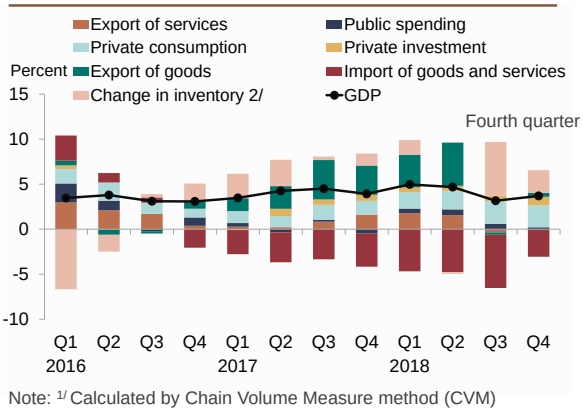


Note: *Average of headline inflation in Indonesia, South Korea, Malaysia, the Philippines, Singapore and Taiwan
Source: CEIC

Thai Economy

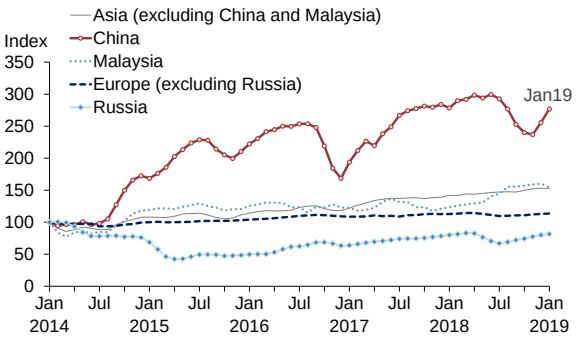
The Thai economy continued to gain traction. Economic growth would be supported by the expansion of private consumption thanks to a more broad-based improvement in household income. Private investment expanded, while public expenditure continued to drive the economy through spending on both consumption and investment. Meanwhile, overall merchandise exports slowed down in tandem with trading partner economies and given the impact from trade protectionist measures between the U.S. and China. In addition, the number of foreign tourists picked up, partly due to the extended exemption of visa-on-arrival (VOA) until the end of April 2019 and new flight routes from China and India.

Contribution to Thailand's GDP growth[1/]



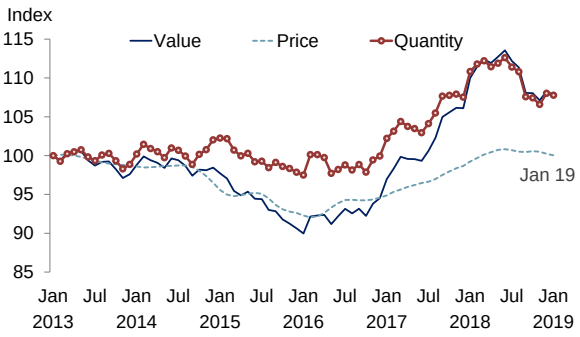
Note: [1/] Calculated by Chain Volume Measure method (CVM)
[2/] Change in inventory and statistical discrepancy
Source: Office of National Economic and Social Development Council, calculations by the Bank of Thailand

Index of foreign tourists classified by nationality
(three-month moving average, seasonally adjusted; January 2014 = 100)



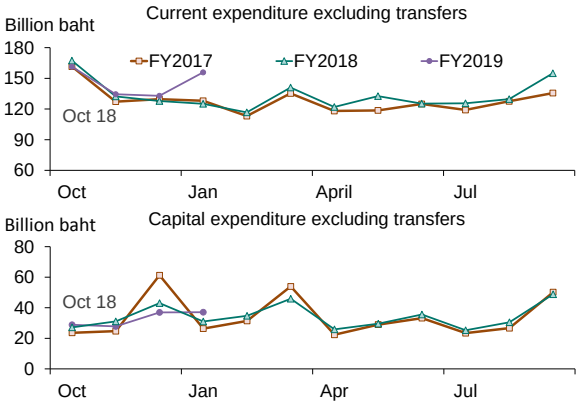
Source: Department of Tourism

Thai exports (excluding gold): value, price and quantity
(3-month moving average, seasonally adjusted; January 2013 = 100)



Source: Customs Department and Ministry of Commerce, calculations by Bank of Thailand

Public spending by central government

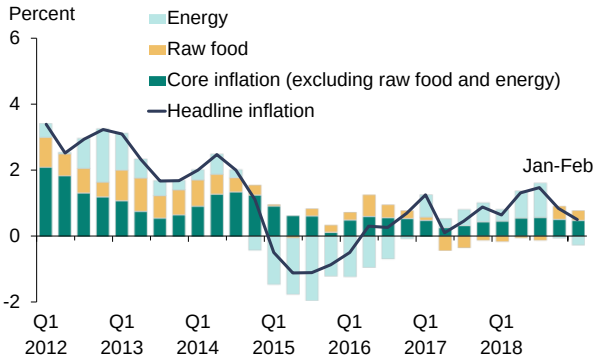


Source: Bureau of Budget, Fiscal Policy Office

Inflation

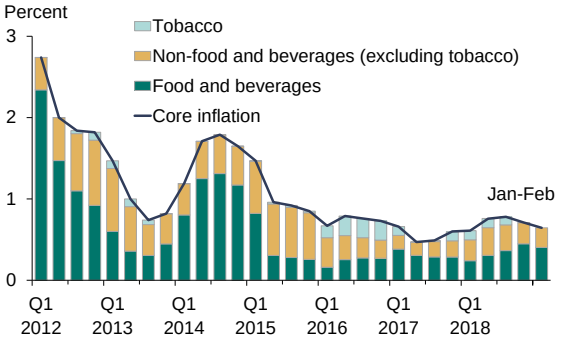
Headline inflation declined mainly due to the fall in energy prices. Core inflation decreased due to slower improvement in prices of processed foods and non-alcoholic beverages, as well as house rents. Nevertheless, structural changes, such as rising e-commerce trends and price competition, would reduce upward pressures on core inflation in non-food categories in the period ahead.

Contribution to headline inflation



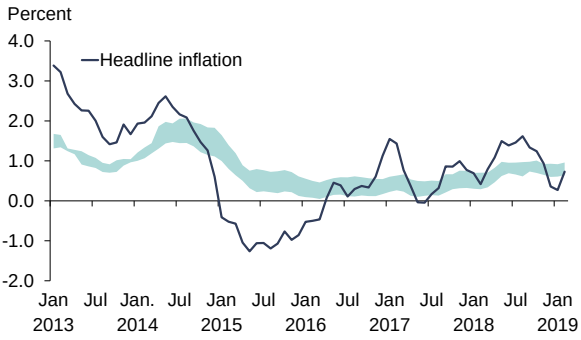
Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

Contribution to core inflation



Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

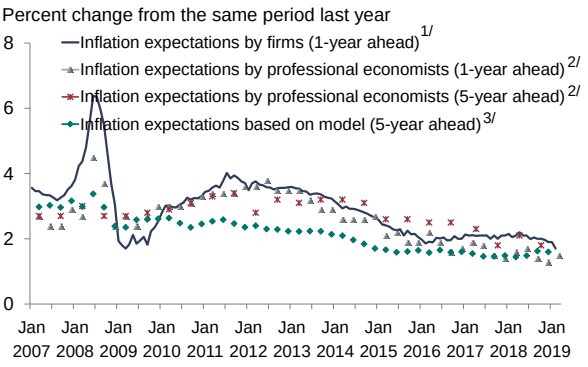
Underlying inflation indicators



Note: The field shows the highest and lowest outcomes among different measures of underlying inflation. The measures included are 1. **Trimmed mean** (excludes goods and services with most volatile price changes, removing the bottom 15 percentile and the top 10 percentile), 2. **Principal component model** (calculates changes in common statistical components that attribute price movements across categories of goods and services) and 3. **Core inflation excluding rents and government measures.**

Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

Inflation expectations

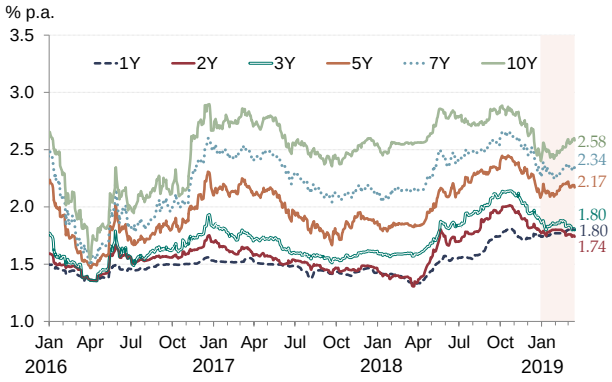


Source: 1/ Business Sentiment Survey (BSI) by the Bank of Thailand
2/ Asia Pacific Consensus Forecast
3/ Calculations based on macro-finance term structure model using bond yield and macroeconomic data

Financial conditions

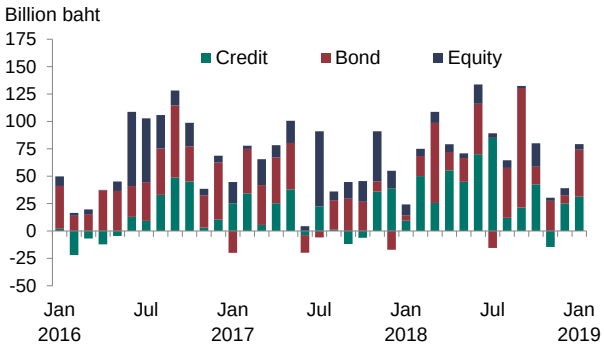
Short-term money market rates and short-term government bond yields gradually rose in line with the policy rate. Meanwhile, medium-term and long-term government bond yields fluctuated owing to both domestic and external factors. Private credit extended to both businesses and households continued to expand. The baht appreciated against the U.S. dollar in line with movements of regional currencies, following a more dovish stance of the Fed's monetary policy communications and less concern of investors over global economic uncertainties. Since the end of February 2019, the baht weakened against the U.S. dollar due to geopolitical risks and Thailand's general election.

Government bond yields



Source: Thai Bond Market Association (Thai BMA) (data as of 19 March 2019)

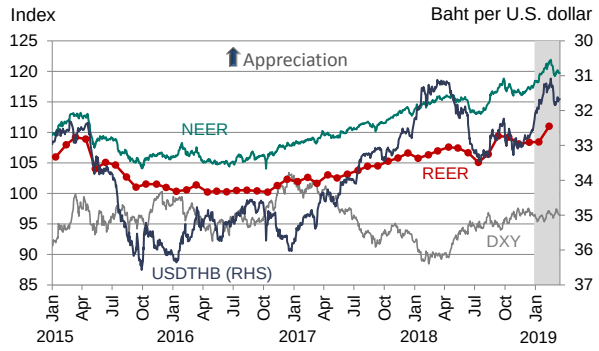
Total corporate financing by instrument*



Note: *Monthly change in outstanding of corporate loans (seasonally adjusted), corporate bonds excluding those issued by commercial banks, and newly issued equities.

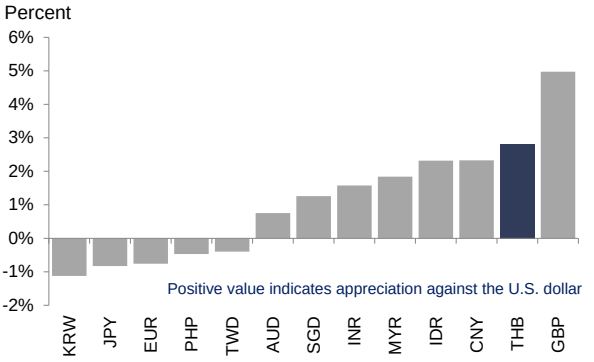
Sources: Bank of Thailand and Thai Bond Market Association (Thai BMA)

The Thai baht vis-a-vis U.S. dollar (USDTHB), Nominal Effective Exchange Rate (NEER), and the Dollar Index (DXY)



Source: Bank of Thailand and Reuters (data as of 19 March 2019)

Currency movements vis-a-vis the U.S. dollar (19 March 2019 compared to 28 December 18)

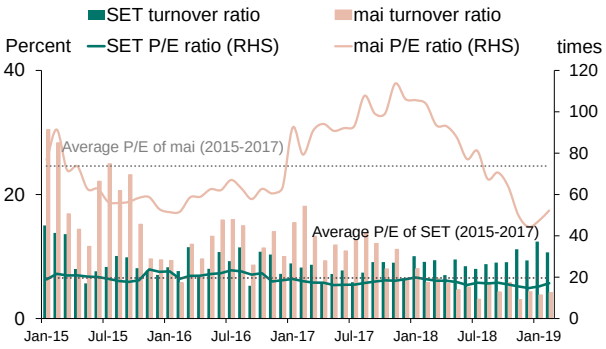


Source: Bank of Thailand and Reuters (data as of 19 March 2019)

Stability: financial markets

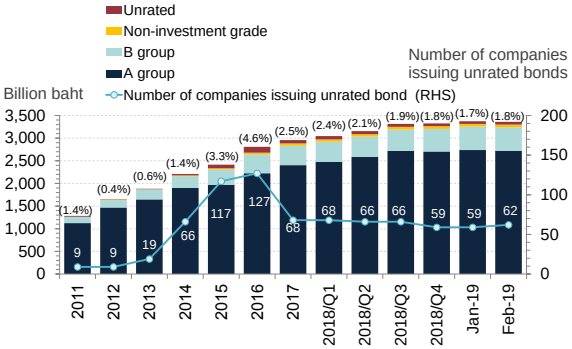
The price-to-earning (P/E) ratio of the Stock Exchange of Thailand (SET) stayed below the historical average, while that of the Market for Alternative Investment (mai) increased. The share of unrated bonds issuance remained stable.

Current price-to-earning ratio and turnover ratio of SET and mai



Source: Stock Exchange of Thailand (as of February 2019)

Corporate bonds outstanding

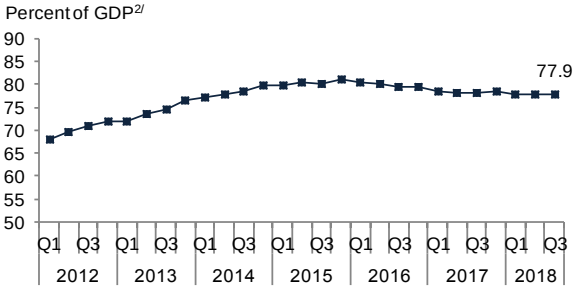


Note: ( ) represents percentage of unrated bonds to total corporate bonds
Source: Thai Bond Market Association (Thai BMA)

Stability: household sector

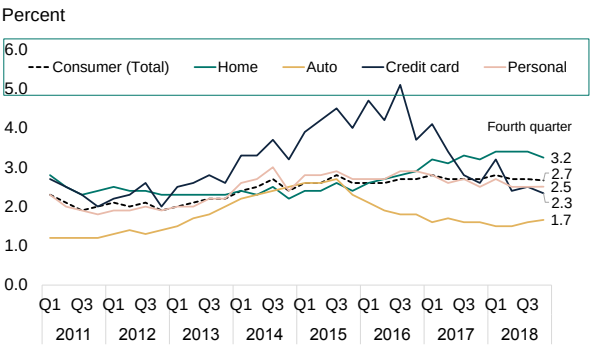
The ratio of household debt to GDP remained high and slightly increased from the previous quarter. Overall credit quality of consumer loans, as indicated by the NPL ratio, remained stable. This was partly a result of debt restructuring and delinquent debt write-offs by financial institutions. However, the NPL ratio of mortgage loans remained high. Meanwhile, credit quality of auto hire purchase loans continued to deteriorate, as reflected by the rising NPL ratio of auto hire purchase loans since the second half of 2018.

Household debt[1/]



Note: [1/] Loans to households by financial institutions
[2/] Calculated by averaging the 4 latest quarterly GDP
[3/] Household debt and GDP data are revised. This results in the different debt to GDP ratios compared to the last MPR.
Source: Bank of Thailand

Share of non-performing loans (NPL) in consumer loans, classified by loan type

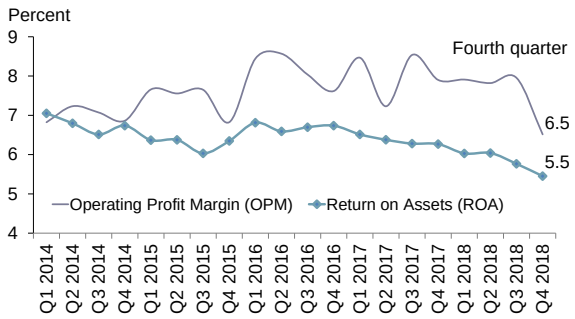


Source: Bank of Thailand

Stability: corporate sector

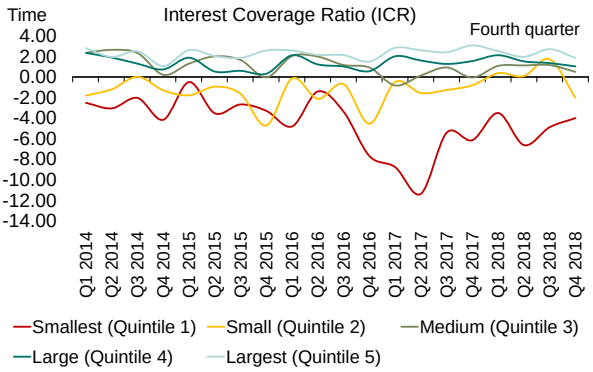
Continued economic expansion benefited financial positions of the corporate sector, especially large corporates with higher profitability. Meanwhile, debt serviceability of SMEs must be monitored, particularly small businesses in certain sectors including commercial and services sectors. Moreover, the NPL ratio among SMEs stood at 4.5 percent, above the historical average.

Operating Profit Margin (OPM) and Return on Assets (ROA)*



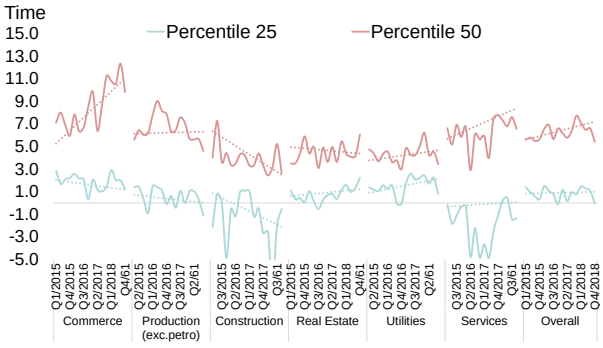
Note: * Median estimates; ROA is returns to average assets.
OPM is operating profits to total sales.
Source: Stock Exchange of Thailand, calculation by Bank of Thailand

Debt serviceability at 25th percentile of each group of firm size



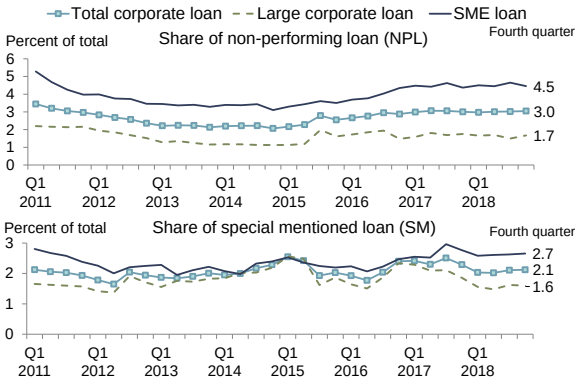
Source: Stock Exchange of Thailand, calculation by Bank of Thailand

Interest Coverage Ratio, classified by sectors



Note: * production exclude Petroleum and chemicals
Source: Stock Exchange of Thailand, calculation by Bank of Thailand

Loan quality of corporate sector

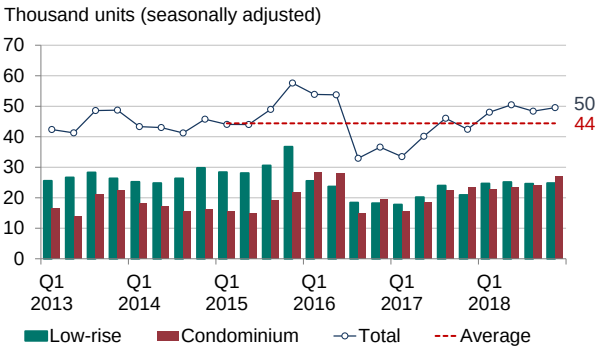


Source: Bank of Thailand

Stability: real estate

Demand for residences continued to rise since the fourth quarter of 2018. This was partly due to acceleration in residential ownership transfers prior to the implementation of macroprudential measure on mortgage loans in April 2019. On the supply side, newly opened residential units expanded after having slowed down in the late 2018. Property prices were broadly unchanged, except prices of condominium.

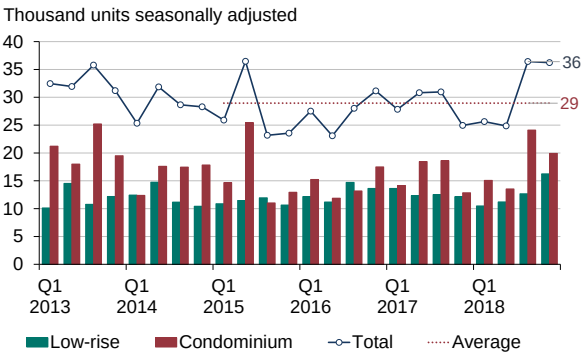
Residential Transfer units in Bangkok and its vicinity



Note: *Average during 2014-2017

Source: Real Estate Information Center

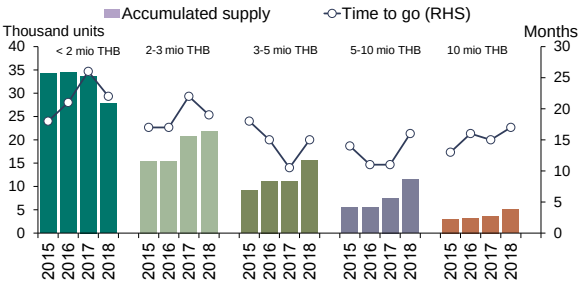
New residential projects launched in Bangkok and its vicinity



Source: Agency for Real Estate Affairs (AREA), calculation by Bank of Thailand

Note: *Average during 2014-2017

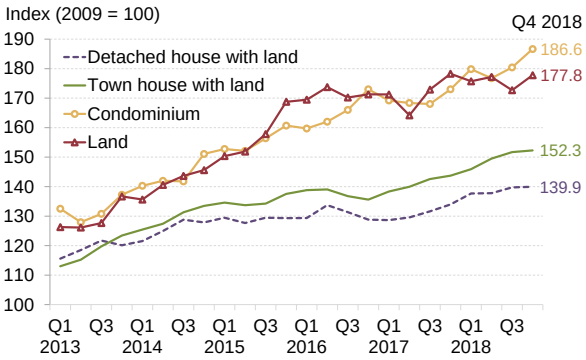
Condominium inventory in Bangkok and vicinity and 'Time to go'



Note: 'Time to go' is the time taken for all real estate inventory to be sold out at the average sales rate per month (since projects launched) given no additional supply.

Source: AREA and calculation by the Bank of Thailand

Real estate price indices



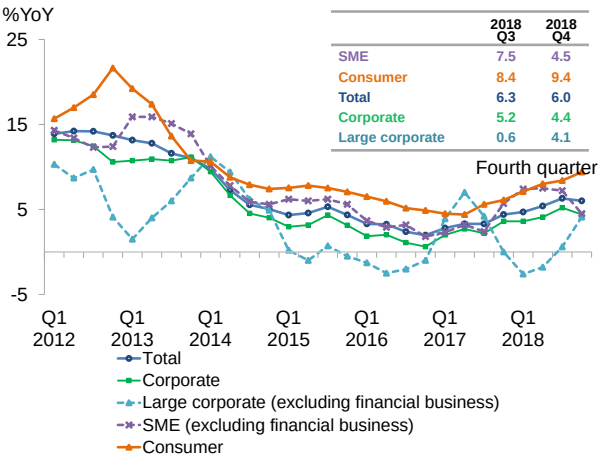
Note: Calculation based on commercial bank loan data

Source: Bank of Thailand

Stability: financial institutions

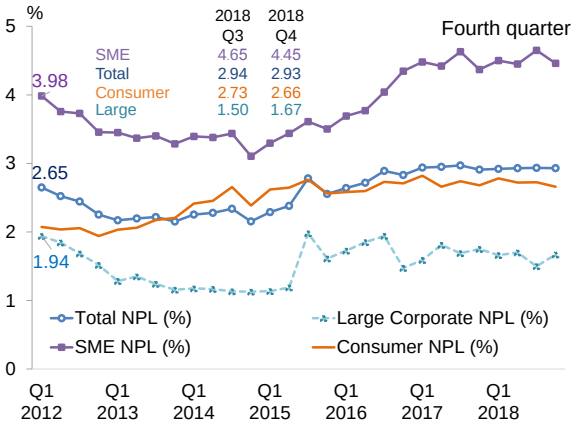
Financial institutions maintained strong financial positions, as reflected in high levels of capital buffers among commercial banks to cushion against risks should credit quality deteriorate. In the fourth quarter of 2018, commercial bank credits continued to expand mainly due to consumer loans. Meanwhile, overall NPL ratio stabilized at a high level, especially for loans extended to SMEs. Meanwhile, the NPL ratio among large corporates increased slightly.

Credit growth in the commercial bank system



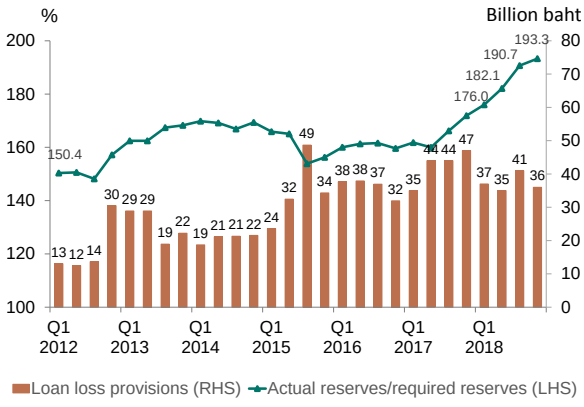
Source: Bank of Thailand

Non-performing loan (NPL)



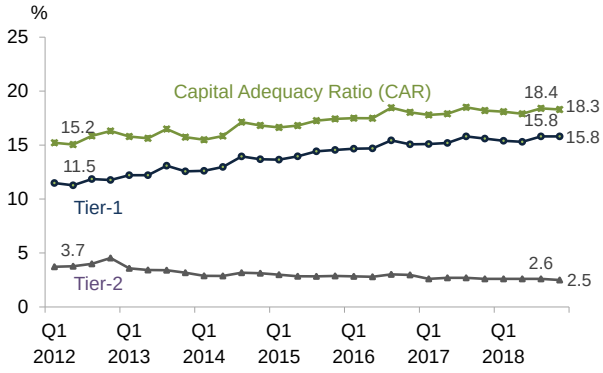
Source: Bank of Thailand

Provisions in commercial bank system



Source: Bank of Thailand

Capital buffers in commercial bank system

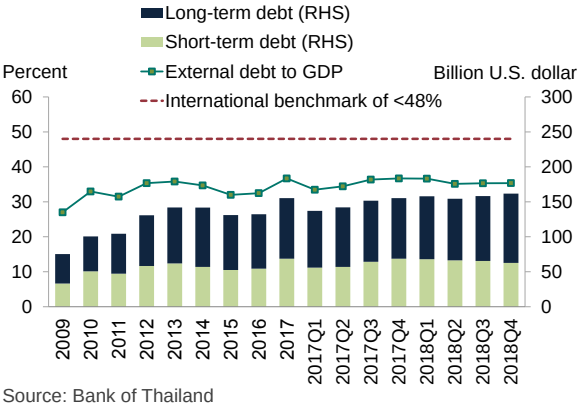


Source: Bank of Thailand

Stability: external position

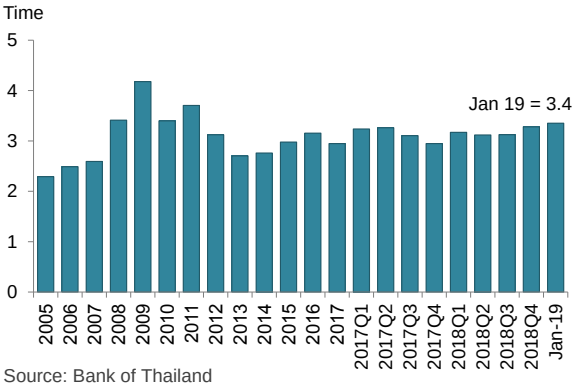
Thailand's external stability remained sound, as reflected in high levels of international reserves and a sustained current account surplus. Moreover, the ratio of external debt to GDP was below an international benchmark. This would help the Thai economy to be resilient against volatilities in global financial market.

Thailand's external debt



Source: Bank of Thailand

Reserve to short-term debt

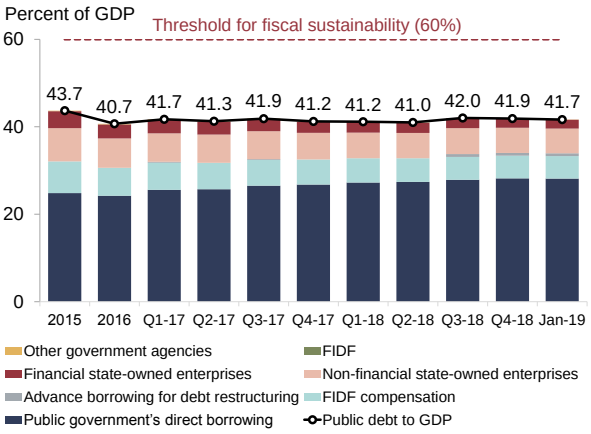


Source: Bank of Thailand

Stability: fiscal sector

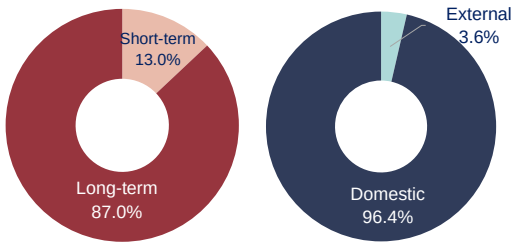
Fiscal stability remained sound. The ratio of public debt to GDP stayed below the sustainability threshold.

Public debt to GDP



Note: Calculated by GDP with Chain Volume Measure
Source: Public Debt Management Office

Outstanding debt as of January 2019



Note: Share of short-term and long-term debt calculated from remaining duration until maturity
Source: Public Debt Management Office


Pursuing Sustainable
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